arXiv:2406.00922v3 [cs.CL] 7 Nov 2024

MEDIQ: Question-Asking LLMs and a Benchmark for Reliable Interactive Clinical Reasoning

Shuyue Stella Li[1] **Vidhisha Balachandran**[2] **Shangbin Feng**[1] **Jonathan S. Ilgen**[1]
Emma Pierson[3] **Pang Wei Koh**[1,4] **Yulia Tsvetkov**[1]
[1]University of Washington [2]Carnegie Mellon University [3]Cornell Tech
[4]Allen Institute for AI
{stelli, yuliats}@cs.washington.edu
https://github.com/stellalisy/mediQ
https://huggingface.co/datasets/stellalisy/mediQ

Abstract

Users typically engage with LLMs interactively, yet most existing benchmarks evaluate them in a static, single-turn format, posing reliability concerns in interactive scenarios. We identify a key obstacle towards reliability: LLMs are trained to answer any question, even with incomplete context or insufficient knowledge. In this paper, we propose to change the static paradigm to an interactive one, develop systems that *proactively ask questions* to gather more information and respond reliably, and introduce an benchmark—**MEDIQ**—to evaluate question-asking ability in LLMs. MEDIQ simulates clinical interactions consisting of a Patient System and an adaptive Expert System; with potentially incomplete initial information, the Expert refrains from making diagnostic decisions when unconfident, and instead elicits missing details via follow-up questions. We provide a pipeline to convert single-turn medical benchmarks into an interactive format. Our results show that directly prompting state-of-the-art LLMs to ask questions *degrades* performance, indicating that adapting LLMs to proactive information-seeking settings is nontrivial. We experiment with abstention strategies to better estimate model confidence and decide when to ask questions, improving diagnostic accuracy by 22.3%; however, performance still lags compared to an (unrealistic in practice) upper bound with complete information upfront. Further analyses show improved interactive performance with filtering irrelevant contexts and reformatting conversations. Overall, we introduce a novel problem towards LLM reliability, an interactive MEDIQ benchmark and a novel question-asking system, and highlight directions to extend LLMs' information-seeking abilities in critical domains.

1 Introduction

General-purpose large language models (LLMs) are designed to serve a broad audience by following instructions and providing the most likely and general answers (Brown et al., 2020; Ouyang et al., 2022). However, in high-stakes decision making scenarios such as clinical conversations, LLM assistants can be harmful if they provide general responses instead of gathering missing information to make informed decisions. As shown in Figure 1, standard medical question-answering (QA) tasks are formulated in a single-turn setup where all necessary information is provided upfront, and the model is not expected to interact with users (Jin et al., 2021; Pal et al., 2022; Jin et al., 2019; Hendrycks et al., 2020). This QA paradigm diverges from real-world scenarios, where users may provide **incomplete information**, and effective decision-making often requires an **investigative process** involving follow-up questions to clarify and gather necessary details (Trimble & Hamilton, 2016; Bornstein & Emler, 2001; Masic, 2022).

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

arXiv:2406.00922v3 [cs.CL] 7 Nov 2024

MEDIQ：提问型大型语言模型及可靠互动临床推理基准

李书阅Stella[1] 维迪莎·巴拉钱德兰[2] 冯尚斌[1] 乔纳森·S·伊尔根[1]艾玛·皮尔森[3] 许鹏伟[1,4] 尤莉娅·茨维特科夫[11]华盛顿大学 [2]卡内基梅隆大学 [3]康奈尔科技[4]艾伦人工智能研究所 {stelli, yuliats}@cs.washington.edu
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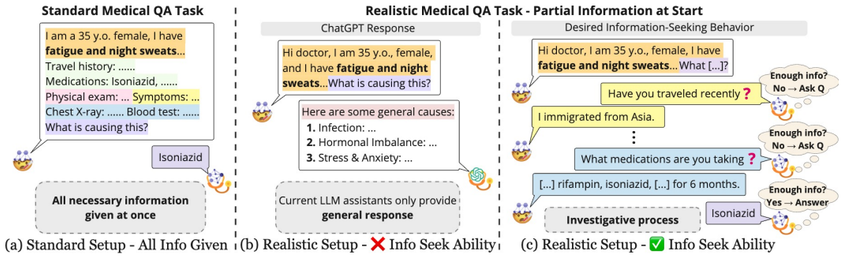
摘要

用户通常以互动方式使用大型语言模型（LLM），然而大多数现有基准测试是在静态、单轮的格式下评估它们，这在互动场景中带来了可靠性问题。我们确定了通向可靠性的一个关键障碍：LLM 被训练为回答任何问题，即使在上下文不完整或知识不足的情况下也是如此。在本文中，我们提出将静态范式改为互动范式，开发能够主动提问以收集更多信息并可靠回应的系统，并引入一个基准测试——MEDIQ——用于评估 LLM 的提问能力。MEDIQ 模拟临床互动，包括一个患者系统和一个自适应专家系统；在初始信息可能不完整的情况下，专家在缺乏信心时不会做出诊断决策，而是通过后续问题获取缺失的细节。我们提供了将单轮医学基准转化为互动格式的流程。我们的结果表明，直接提示最先进的

1 引言

通用大型语言模型（LLM）旨在通过遵循指令并提供最可能和最通用的答案来服务广泛的用户群体（Brown 等，2020；Ouyang 等，2022）。然而，在高风险决策场景中，如临床对话，如果 LLM 助手提供的是一般性回答而不是收集缺失信息以做出明智决策，可能会造成伤害。如图 1 所示，标准的医学问答（QA）任务通常是在单轮设置中制定的，其中所有必要信息在一开始就已提供，模型不需要与用户互动（Jin 等，2021；Pal 等，2022；Jin 等，2019；Hendrycks 等，2020）。这种 QA 模式与现实场景存在差异，在现实中，用户可能提供不完整的信息，而有效的决策通常需要一个调查过程，包括通过后续问题来澄清和收集必要的细节（Trimble & Hamilton，2016；Bornstein & Emler，20

第38届神经信息处理系统大会（NeurIPS 2024）。



(a) Standard Setup - All Info Given (b) Realistic Setup - ❌ Info Seek Ability (c) Realistic Setup - ✅ Info Seek Ability

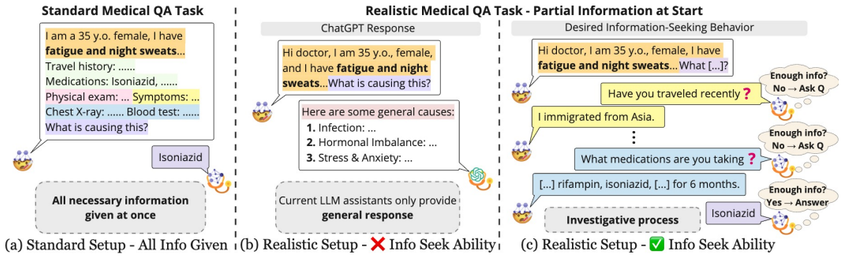
Figure 1: Information Seeking Task. In standard medical QA tasks (left), all necessary information is given to the assistant model at the same time. When given partial information, current LLMs only provides general responses (middle). In a more realistic scenario (right), the presentation of patient information relies on proactive elicitation from the doctor; our proposed MEDIQ framework operationalizes this scenario.

This gap between existing benchmarks and reality calls for a paradigm shift to designing systems adept at navigating high-stakes interactive scenarios. Focusing on clinical interactions where context is often incomplete, we introduce **MEDIQ**, an interactive benchmark for **m**edical **e**valuation with **d**ynamic **i**nformation-seeking **q**uestions, to address limitations of static single-turn QA benchmarks (Figure 2). Unlike conventional systems, which assume that all necessary information is readily available, MEDIQ acknowledges the inherent uncertainty in medical consultations where a typical patient does not have the expertise to distill all necessary and relevant information they need to provide. To achieve this, MEDIQ comprises two components: a **Patient system** that simulates a patient and responds to follow-up questions, and an **Expert system** that serves as a doctor's assistant and asks questions to the patient before making a medical decision. In this **interactive clinical reasoning task**, a successful information-seeking Expert should decide, at each turn, whether it has enough information to provide a confident answer; if not, it should ask a follow-up question.

We convert two medical QA datasets, MEDQA (Jin et al., 2021) and CRAFT-MD (Johri et al., 2023, 2024), into an interactive benchmark by parsing the patient records to only provide partial information in the beginning. We first develop and validate a Patient system that accurately answers Expert inquiries by retrieving the correct facts from the patient record. We then benchmark Expert systems based on state-of-the-art (SOTA) LLMs, including Llama-3 (Touvron et al., 2023), GPT-3.5 (Brown et al., 2020) and GPT-4 (OpenAI et al., 2024), to evaluate their proactive information seeking ability. It is striking that prompting these models to ask questions results in an 11.3% accuracy drop compared to starting with the same limited information and asking no questions, showing that adapting LLMs to interactive information-seeking settings is nontrivial. A key challenge is deciding when to ask a follow-up question instead of directly providing an answer. With confidence estimation strategies such as rationale generation and self-consistency, we improve Expert performance by 22.3%, although a 10.3% gap remains compared to an upper bound when full information is presented at once.

Our results show that while SOTA LLMs perform relatively well with complete information, they struggle to proactively seek missing information in a more realistic, interactive settings with incomplete initial information. By providing a modular, interactive benchmark, we hope to facilitate the development of reliable LLM assistants for complex decision-making in healthcare and other high-stakes domains. Our main contributions are:

1. We identify the critical problem of **information-seeking questions** in reliable interactive LLM assistants. We propose a paradigm shift and a practical conversion pipeline from standard single-turn benchmarks into interactive settings with incomplete initial information.
2. We develop the **MEDIQ Benchmark** to simulate more realistic clinical interactions between a Patient System and an Expert System. We rigorously develop and test the Patient System to benchmark any Expert's information-seeking and clinical decision-making abilities.
3. We show that SOTA LLMs such as Llama-3-Instruct, GPT-3.5 and GPT-4 struggle at proactive information seeking, revealing a significant gap in this area.
4. We propose **MEDIQ-Expert**, our best Expert system with novel abstaining capabilities to reduce unconfident answers, to partially close the gap between the more realistic incomplete information setup and the existing full information setup.



(a) Standard Setup - All Info Given (b) Realistic Setup - ❌ Info Seek Ability (c) Realistic Setup - ✅ Info Seek Ability

图 1：信息搜索任务。在标准的医学问答任务中（左图），所有必要的信息会同时提供给助手模型。当只给出部分信息时，目前的大型语言模型仅提供一般性回答（中图）。在更现实的场景中（右图），患者信息的呈现依赖于医生的主动引导；我们提出的 MEDIQ 框架将此场景操作化。

现有基准与现实之间的差距呼唤在设计能够处理高风险互动场景的系统方面进行范式转变。针对情境往往不完整的临床互动，我们引入了MEDIQ，一种用于医学评估的互动基准，具有动态信息获取问题，以解决静态单轮问答基准的局限性（图2）。与假设所有必要信息都唾手可得的传统系统不同，MEDIQ承认医学咨询中固有的不确定性，因为典型患者没有能力提炼出他们需要提供的所有必要和相关信息。为实现此目标，MEDIQ包括两个组件：一个模拟患者并对跟进问题作出回答的患者系统，以及一个作为医生助手向患者提问然后做出医疗决策的专家系统。在这一互动临床推理任务中，成功的

我们将两个医学问答数据集，MEDQA（Jin 等，2021）和 CRAFT-MD（Johri 等，2023，2024），通过解析患者记录以在开始时仅提供部分信息，转换为一个交互式基准。我们首先开发并验证了一个患者系统，该系统通过从患者记录中检索正确事实，能够准确回答专家的询问。然后，我们基于最先进（SOTA）的 LLM，包括 Llama-3（Touvron 等，2023）、GPT-3.5（Brown 等，2020）和 GPT-4（OpenAI 等，2024），对专家系统进行基准测试，以评估其主动获取信息的能力。令人惊讶的是，提示这些模型提出问题比在相同有限信息下不提问开始时的准确率下降了 11.3%，这表明将 LLM 适配到交互式信息获取环境是非平凡的。一个关键挑战是确定何时提出后续问题而不是直接提供答案。通过信心估计策略如

我们的结果表明，虽然最先进的大型语言模型（SOTA LLMs）在信息完整时表现相对良好，但在更现实的、初始信息不完整的互动环境中，它们在主动寻找缺失信息方面表现不佳。通过提供一个模块化、互动的基准，我们希望促进在医疗和其他高风险领域中开发可靠的大型语言模型助理，以支持复杂决策。我们的主要贡献如下：

1. 我们识别了可靠互动式大型语言模型助手中信息获取问题的关键问题。我们提出了一种范式转变，以及将标准单轮基准测试转换为具有不完整初始信息的互动设置的实际转换流程。 2. 我们开发了 MEDIQ 基准，以模拟患者系统与专家系统之间更现实的临床互动。我们严格开发和测试患者系统，以评估任何专家的信息获取能力和临床决策能力。 3. 我们展示了像 Llama-3-Instruct、GPT-3.5 和 GPT-4 这样的最先进大型语言模型在主动信息获取方面表现不佳，揭示了该领域的显著差距。 4. 我们提出了 MEDIQ-Expert，这是我们最优的专家系统，具备新颖的放弃能力以减少不自信的回答，从而部分弥补更现实的不完整信息设置与现有完整信息设置之间的差距。

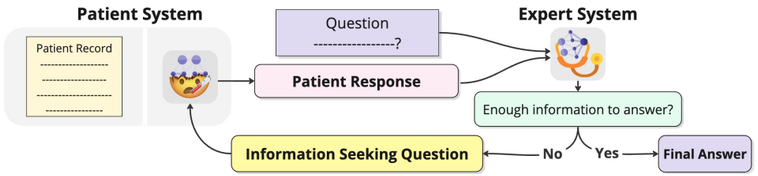


Figure 2: The MEDIQ Benchmark. MEDIQ operationalizes a more realistic dynamic clinical interaction between a Patient system and an Expert system to evaluate info-seeking and question-asking.

2 MEDIQ: Dynamic Medical Consultation Framework Overview

Task Definition The dynamic medical consultation task simulates the iterative nature of real-world clinical interactions. This task starts by providing an initial patient description k_0 of their conditions to the Expert system. The initial information typically contains the patient's age, gender, and chief complaint for the visit. The Patient system has access to the entire patient record $\mathcal{K} = \{k_0, k_1, \ldots, k_n\}$, and the necessary information to answer the multiple choice question is $\mathcal{K}^* \subseteq \mathcal{K}$. At the start of the t-th turn, the knowledge available to the Expert system is denoted as $\mathcal{K}_{t-1} = \{k_0, \ldots, k_i\}$. Given follow-up question q_t, the Patient system responds with $r_t = \{k | k \in \mathcal{K}\}$. The Expert knowledge is then updated as $\mathcal{K}_t = \mathcal{K}_{t-1} \cup r_t$. The **main challenge of the task** is for the Expert system to ask information-seeking questions to expand $\mathcal{K}_t$ until the knowledge gap is filled, i.e. $\mathcal{K}_t = \mathcal{K}^*$, at which point the Expert system is asked to make a final decision.

2.1 The Patient System

Patient Task As part of the MEDIQ framework, the Patient system simulates a human patient in clinical conversations. The Patient system has access to the full patient record that is sufficient for the diagnosis, including symptoms, onset duration, medical history, family history, and/or relevant lifestyle factors. The Patient system uses the patient record and a single information-seeking question from the Expert system to produce a coherent response consistent with the given patient information as shown in Figure 2. A reliable Patient system is critical to simulate a real and accurate medical consultation process. We propose that any Patient system should be evaluated on (1) *Factuality* - measuring if a patient's responses are faithful to the patient's record and history and (2) *Relevance* - measuring if the patient's response answers the expert's question. Given the full patient record and the expert question, we propose and evaluate three Patient system variants: **Direct**, **Instruct**, and **Fact-Select**, to obtain the patient response. Exact prompts and examples are in Appendix A.2.

1. **Direct:** Serving as a baseline, the Patient treats the response-generation as a reading comprehension task with no additional instruction. The prompt includes the patient's record followed by the Expert's question and asks the model to directly respond to the question using the given paragraph.
2. **Instruct:** The Patient is instructed to respond truthfully to the Expert's question using the patient record only. When the context does not contain an answer to the question, the Patient is instructed to refrain from answering.
3. **Fact-Select:** The Patient aims to improve the factuality of the response by decomposing the patient record into atomic facts and responds by selecting facts that are relevant to the Expert's question.

2.2 The Expert System

Expert Task The Expert system simulates the medical decision-making process of experienced clinicians, who seek additional patient information and iteratively update their differential diagnosis. The Expert system is first presented with a medical question and limited patient information. As each turn, it assesses whether the provided information is sufficient to answer the question. If the Expert system is unconfident, it can elicit evidence with a follow-up information-seeking question; otherwise, the Expert system deems the acquired information sufficient and provides a final answer. The performance of the Expert system is evaluated on the (1) *efficiency* of the conversation (number of follow-up questions) and (2) the *accuracy* of the final diagnosis.

2.2.1 Expert System Breakdown

Medical decision making is a complex process involving clinical reasoning and proactive information-seeking (Bordage, 1999; Norman, 2005; Schmidt et al., 1990; Boshuizen & Schmidt, 1992; Patel

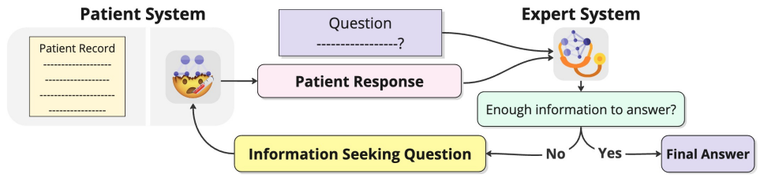


图2：MEDIQ基准。MEDIQ实现了患者系统与专家系统之间更现实的动态临床交互，以评估信息寻求和提问行为。

2 MEDIQ：动态医疗咨询框架概述

任务定义 动态医疗咨询任务模拟了真实世界临床互动的迭代性质。该任务从向专家系统提供患者病情的初步描述 k_0 开始。初步信息通常包含患者的年龄、性别以及此次就诊的主诉。患者系统可以访问完整的病历记录 $\mathcal{K} = \{k_0, k_1, \ldots, k_n\}$，并且回答多项选择题所需的信息是 $\mathcal{K}^* \subseteq \mathcal{K}$。在第 t 轮开始时，专家系统可用的知识表示为 $\mathcal{K}_{t-1} = \{k_0, \ldots, k_i\}$。在接到后续问题 q_t 后，患者系统会以 $r_t = \{k | k \in \mathcal{K}\}$ 作出回应。随后，专家知识更新为 $\mathcal{K}_t = \mathcal{K}_{t-1} \cup r_t$。该任务的主要挑战在于专家系统需要提问以获取信息，从而扩展 $\mathcal{K}_t$ 直到填补知识空白，即 $\mathcal{K}_t = \mathcal{K}^*$，此时要求专家系统做出最终决定。

2.1 患者系统

患者任务 作为MEDIQ框架的一部分，患者系统在临床对话中模拟真人患者。患者系统可以访问完整的患者记录，这些记录足以用于诊断，包括症状、发病时间、病史、家族史和/或相关的生活方式因素。患者系统使用患者记录和来自专家系统的单个信息查询问题，生成与给定患者信息一致的连贯回应，如图2所示。可靠的患者系统对于模拟真实且准确的医疗咨询过程至关重要。我们建议对任何患者系统进行评估时应考虑以下两点：(1) 真实性——衡量患者的回应是否忠实于患者的记录和病史；(2) 相关性——衡量患者的回应是否回答了专家的问题。在给定完整患者记录和专家问题的情况下，我们提出并评估了三种患者系统变体：直接型、指导型和

1. 直接：作为基线，患者将响应生成视为一个阅读理解任务，没有额外的指令。提示包括患者的记录，随后是专家的问题，并要求模型直接使用给定段落回答问题。

2. 指示：患者被指示仅使用患者记录如实回答专家的问题。当上下文中没有包含问题的答案时，患者被指示避免回答。3. F

act-select：患者旨在通过分解患者来提高回应的真实性
将信息分解为原子事实，并通过选择与专家问题相关的事实来响应 n.

2.2 专家系统

专家任务 专家系统模拟经验丰富的临床医生的医疗决策过程，这些医生会寻求额外的患者信息并迭代更新他们的鉴别诊断。专家系统首先会收到一个医疗问题和有限的患者信息。在每一轮中，它会评估提供的信息是否足以回答该问题。如果专家系统没有信心，它可以通过后续的信息获取问题来获取证据；否则，专家系统认为获得的信息足够并提供最终答案。专家系统的表现评估标准包括：(1) 对话的效率（后续问题的数量）和 (2) 最终诊断的准确性。

2.2.1 专家系统分解

医疗决策是一个复杂的过程，涉及临床推理和主动的信息寻求（Bordage, 1999；Norman, 2005；Schmidt 等, 1990；Boshuizen & Schmidt, 1992；Patel

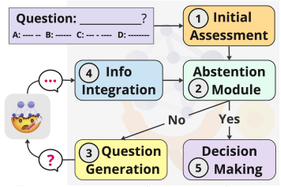


Figure 3: Expert system information flow breakdown.

et al., 1994). We describe our proposed **MEDIQ-Expert**, which operationalizes the Expert system by breaking down the task into five medically-grounded steps: (1) initial assessment, (2) abstention, (3) question generation, (4) information integration, and (5) decision making (Figure 3). Each step is modular and easily modifiable.

Step 1. Initial Assessment Module: Given limited patient intake information and the multiple choice question (MCQ) as the input, the goal of this module is to provide an initial assessment of the patient. The Expert system is asked to produce a paragraph that elaborates on the symptoms and options, and identifying potential knowledge gaps (e.g., additional symptoms, lab tests) missing for answering the question. This step is done only once at the beginning of the interaction, and we keep the output in the conversation thread for future turns to refer back to.

Step 2. Abstention Module: When the model is not confident, it should *abstain* from giving an answer and asks a information-seeking question instead. The goal of the Abstention Module is to evaluate the **confidence level** of the Expert system to make a decision given the available information. The input to this module is the MCQ and the patient information consisting of the initial presentation and a conversation log of follow-up questions and responses. We probe the confidence level of the model to reliably answer the question via prompting (§ 2.2.2). The output of this module is a yes/no answer for whether to proceed to final answer. If the model is confident, it skips to decision making; otherwise, it continues to question generation.

Step 3. Question Generation Module: When more information is deemed necessary, the goal of the question generation module is to craft an information-seeking question to elicit additional medical evidence such as lifestyle factors and physical exam results. The input to this module is all previous reasoning steps, and the acquired patient information; the notion of atomic questions is defined with respect to the medical domain in the prompt, and the output is an atomic question to the patient.

Step 4. Information Integration Module: When a patient response is returned to the Expert system, the information integration module aggregates all gathered patient information up to this point to update the understanding of the patient condition. This step simply appends a question-answer pair to the end of an existing conversation log, which will then be passed to the Abstention Module.

Step 5. Decision Making Module: When enough evidence is gathered, the Expert system leverages integrated patient information and medical knowledge to provide an accurate answer to the question. The input to this module is previous reasoning steps, the MCQ, and the gathered patient information, and the output is the chosen option. Exact prompts for all above sections are in Appendix B.

2.2.2 Expert System Variants with Different Abstention Strategies

One component of active information-seeking is the ability to decide *when* to ask questions, which we operationalize with the Abstention Module to either *ask* or *answer* at each turn. Abstention reduces LLM hallucinations in low-confidence scenarios (Umapathi et al., 2023; Rawte et al., 2023) and mitigates misleading or insufficiently substantiated conclusions (Feng et al., 2024). We develop the following variants of the Abstention Module via different instructions to the LLM to probe its confidence in whether their parametric knowledge is sufficient to reliably answer the MCQ. Exact prompts are in Appendix B.2.

0. **BASIC:** As a baseline, the model is asked to implicitly indicate its abstain decision by either generating an atomic question or producing an answer to the MCQ.
1. **Numerical:** To get an explicit understanding of the model's confidence, we first prompt the model to generate a numerical confidence score between 0 and 1 following (Tian et al., 2023). Then, an arbitrary threshold is set to either proceed with a final answer or ask a question.

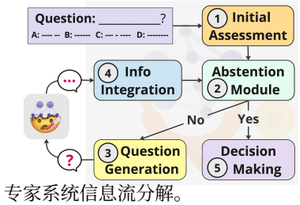


图3：专家系统信息流分解。

et al., 1994)。我们描述了我们提出的 MEDIQ-Expert，它通过将任务分解为五个医学基础步骤来实现专家系统的操作：(1) 初步评估，(2) 回避，(3) 问题生成，(4) 信息整合，(5) 决策（图 3）。每一步都是模块化的且易于修改。

第1步。初步评估模块：在有限的患者接收信息和多项选择题（MCQ）作为输入的情况下，该模块的目标是提供对患者的初步评估。专家系统被要求生成一个段落，详细说明症状和选项，并识别回答问题所缺失的潜在知识空白（例如，额外的症状、实验室检测）。此步骤仅在交互开始时进行一次，并将输出保存在对话线程中，以供未来回合参考。

步骤 2. 放弃模块：当模型没有信心时，它应该放弃给出答案，而是提出一个寻求信息的问题。放弃模块的目标是评估专家系统在给定可用信息的情况下做出决策的信心水平。该模块的输入是多项选择题（MCQ）和患者信息，包括初次表现以及随访问题和响应的对话记录。我们通过提示（§ 2.2.2）来探测模型可靠回答问题的信心水平。该模块的输出是一个是否继续最终回答的是/否答案。如果模型有信心，它将跳到决策阶段；否则，它将继续生成问题。

第三步：问题生成模块：当认为需要更多信息时，问题生成模块的目标是设计一个寻求信息的问题，以获取额外的医疗证据，例如生活方式因素和体检结果。该模块的输入包括所有先前的推理步骤和已获得的患者信息；原子问题的概念在提示中相对于医疗领域定义，输出是向患者提出的一个原子问题。

第4步 信息整合模块：当患者的回应返回到专家系统时，信息整合模块会汇总到目前为止收集的所有患者信息，以更新对患者病情的理解。这一步仅仅是在现有对话记录的末尾添加一个问答对，然后将其传递给回避模块。

第5步 决策模块：当收集到足够的证据时，专家系统利用整合的患者信息和医学知识对问题提供准确的答案。该模块的输入是之前的推理步骤、多项选择题（MCQ）以及收集到的患者信息，输出是所选择的选项。上述所有部分的具体提示见附录B。

2.2.2 具有不同弃权策略的专家系统变体

积极信息搜寻的一个组成部分是决定何时提问的能力，我们通过放弃模块来操作化这一点，使其在每轮中决定提问或回答。放弃模块可以在低置信度情境下减少大型语言模型（LLM）的幻觉（Umapathi 等，2023；Rawte 等，2023），并缓解误导性或证据不足的结论（Feng 等，2024）。我们通过向大型语言模型提供不同的指令，开发了放弃模块的以下变体，以探测其是否有足够的参数知识可靠地回答多项选择题（MCQ）。具体提示见附录 B.2。

0. 基础：作为基线，要求模型通过生成一个原子问题或提供选择题的答案来隐式表示其放弃决策。

1. 数值方法：为了明确理解模型的置信度，我们首先提示模型生成一个介于 0 到 1 之间的数值置信度分数，遵循 (Tian et al., 2023)。然后，设置任意阈值以决定是继续给出最终答案还是提出一个问题。

2. **Binary:** Previous work has shown that LLMs struggle at producing numerical confidence scores (Srivastava et al., 2022; Geng et al., 2023). To address this, the Binary variant enables a simple classification of whether enough information is present. This setup simplifies the decision process, but may lack the nuanced understanding of confidence levels.
3. **Scale:** Binary classification does not provide granularity where the decision is ambiguous. Scale abstention solves this issue by combining direct quantification with a manageable set of discrete, interpretable options. The model is given definitions of confidence levels on a 5-point Likert scale (e.g., "`Very Confident`", "`Somewhat Confident`"), and is asked to select a rating to express its confidence. An arbitrary threshold is set to either proceed with a final answer or ask a question.
4. **Rationale Generation (RG):** Model performance is shown to improve when prompted to generate a reasoning chain about the decision process (Wei et al., 2022; Marasović et al., 2021). This gives the model a longer context window for reasoning, allowing the final decision to be conditioned on previous generations. We attempt to generalize this finding to the more complex interactive medical information-seeking setup by applying it to Numerical, Binary and Scale abstention prompts.
5. **Self-Consistency (SC)**. To further improve the Expert system's abstaining decision, we apply Self-Consistency to the above variants. Self-consistency repeatedly prompts the LLM n times and take the average (Numerical and Scale) or the mode (Binary) of the output as the final output, and is shown to improve model performance (Wang et al., 2022).

3 Experiments

We conduct experiments to validate each component of MEDIQ. First, we evaluate the Patient system with *factuality* and *relevance* metrics (§ 3.1). Then, we establish the correlation between information availability and accuracy by studying model performance with varying levels of input information (§ 3.2.1). Finally, we improve the information-seeking ability of LLMs under MEDIQ (§ 3.2.2).

Evaluation Dataset We convert MEDQA (CC-BY 4.0) (Jin et al., 2021) and CRAFT-MD (CC-BY 4.0) (Johri et al., 2023, 2024) into an interactive setup for our experiments. MEDQA is a standard benchmark for medical question answering with 10178/1272/1273 train/dev/test samples. Each sample contains a paragraph of patient record ending with a multiple choice question. CRAFT-MD contains 140 dermatology patient records in a similar format, among which 100 are collected from an online question bank and 40 are created by expert clinicians. We parse each patient record into age, gender, the chief complaint (primary reason for the clinical visit), and additional evidence. Only the age, gender, and chief complaint are presented to the Expert system, from which it is expected to elicit missing information. The resulting tasks are called iMEDQA and iCRAFT-MD, respectively. See Appendix C for detail.

3.1 Patient System Reliability Evaluation

We automate the evaluation of patient responses with factuality score and relevance score for the ease of scalability, and conduct manual annotations to validate our metrics (Appendix A.4).

Factuality Score measures whether the Patient system's response is consistent with the patient record. Each Patient response is first decomposed into a list of atomic statements, then we compute the percentage of atomic statements that are supported by the information in the patient record. The factuality score is the percent of supported statements averaged over all patients.

Relevance Score measures whether the Patient system's response answers the Expert's question. Since there is no oracle data on the correct answer for Expert follow-up questions, we construct a synthetic parallel evaluation dataset of questions and responses to evaluate the relevance of Patient responses: given a patient record decomposed into atomic statements, we rephrase each statement into an atomic question, for which the statement is the ground truth answer. Then, the Patient system produces a response using the patient record and the generated atomic question. The average embedding semantic similarity between the generated response and the ground truth statement over the evaluation dataset is the resulting relevance score. See Appendix A.1 for more detail.

Setup We use GPT-3.5 as the base LLM for all three variants (Direct, Instruct, and Fact-Select) and compare the factuality and relevance scores. For factuality, we sample 1272 patient cases from MEDIQ interactions with follow-up questions generated by different Expert systems so the Patient

2. 二元：之前的研究表明，大型语言模型在生成数值置信度分数方面存在困难（Srivastava 等，2022；Geng 等，2023）。为了解决这个问题，二元变体允许对信息是否充足进行简单分类。这种设置简化了决策过程，但可能缺乏对置信度水平的细致理解。
3. 规模：二分类在决策不明确时无法提供细化。规模弃权通过将直接量化与一组可管理的离散、可解释选项结合起来解决了这个问题。模型会获得一个 5 点李克特量表上置信水平的定义（例如，“非常自信”、“有些自信”），并被要求选择一个等级来表达其置信度。设置一个任意阈值，以决定是继续给出最终答案还是提出问题。
4. 理由生成(RG)：研究表明，当模型被提示生成关于决策过程的推理链时，其性能会有所提升（Wei 等, 2022；Marasović 等, 2021）。这为模型提供了更长的上下文窗口以进行推理，从而使最终决策可以以之前的生成结果为条件。我们尝试将这一发现推广到更复杂的互动式医疗信息获取场景中，通过将其应用于数值型、二值型和量表型放弃提示。
5. 自一致性（SC）。为了进一步改进专家系统的回避决策，我们对上述变体应用自一致性。自一致性会重复提示大型语言模型（LLM）n 次，并将输出的平均值（数值型和量表型）或众数（二值型）作为最终输出，研究表明这可以提升模型性能（Wang 等，2022）。

3 个实验

我们进行实验以验证MEDIQ的每个组件。首先，我们使用真实性和相关性指标评估患者系统（§ 3.1）。然后，通过研究模型在不同输入信息水平下的表现，我们建立了信息可用性与准确性之间的相关性（§ 3.2.1）。最后，我们在MEDIQ下提高了大型语言模型的信息寻求能力（§ 3.2.2）。

评估数据集 我们将 MEDQA（CC-BY 4.0）（Jin 等，2021）和 CRAFT-MD（CC-BY 4.0）（Johri 等，2023, 2024）转换为交互式实验设置。MEDQA 是医学问答的标准基准，包含 10178/1272/1273 条训练/验证/测试样本。每个样本包含一段以选择题结尾的病人记录。CRAFT-MD 包含 140 条皮肤科病人记录，格式类似，其中 100 条来自在线题库，40 条由专业临床医生创建。我们将每条病人记录解析为年龄、性别、主诉（就诊的主要原因）和附加证据。只有年龄、性别和主诉会呈现给专家系统，系统需要从中引出缺失的信息。由此产生的任务分别称为 iMEDQA 和 iCRAFT-MD。详情见附录 C。

3.1 患者系统可靠性评估

我们通过事实性评分和相关性评分自动评估患者的回应，以便于扩展，并进行人工注释以验证我们的指标（附录A.4）。

事实性评分衡量患者系统的响应是否与患者记录一致。每个患者响应首先被分解为一个原子陈述列表，然后我们计算由患者记录中的信息支持的原子陈述的百分比。事实性评分是对所有患者的支持陈述百分比取平均值。

相关性评分衡量患者系统的响应是否回答了专家的问题。由于没有关于专家后续问题正确答案的权威数据，我们构建了一个问题和响应的合成平行评估数据集来评估患者响应的相关性：给定分解为原子陈述的患者记录，我们将每个陈述重新表述为一个原子问题，该陈述即为该问题的真实答案。然后，患者系统使用患者记录和生成的原子问题生成响应。在评估数据集中，生成的响应与真实陈述之间的平均嵌入语义相似度即为最终的相关性评分。更多细节请参见附录 A.1。

设置 我们使用 GPT-3.5 作为三种变体（直接、指令和事实选择）的基础大语言模型，并比较事实性和相关性评分。对于事实性，我们从 MEDIQ 交互中抽取了 1272 个患者案例，并使用不同专家系统生成的后续问题，以便患者

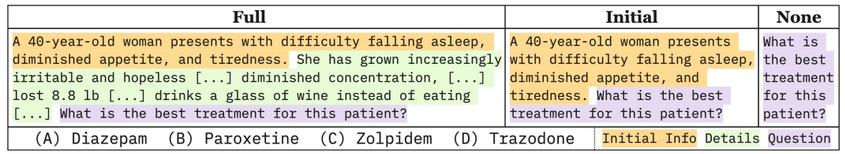


Figure 4: Non-interactive Expert system evaluation at various information availability levels. The question and options are provided to the Expert model in all three settings.

system sees diverse Expert questions and compute the average across all generated questions. For relevance, we use all 1272 patient records from the development set of MEDQA.

3.2 Expert System Experiments

3.2.1 Benchmarking Existing LLMs in Incomplete Information Scenarios

We evaluate the performance of non-interactive Expert systems with varying information availability levels to observe the relationship between information availability and accuracy and to establish baselines. The baselines are evaluated at three initial information availability levels (Figure 4): **Full**, **Initial**, and **None**. The **Full** setup is equivalent to the standard QA task, wherein all patient information is provided to the Expert system in the beginning; **Initial** only discloses the gender, age, and the chief complaint that leads to the clinical visit (e.g. fever, headache, etc.); **None** provides no patient information but only the MCQ to the Expert system.

3.2.2 Interactive Expert Systems

Expert Variants Without explicitly providing the option to ask follow-up questions, vanilla LLMs always answer with incomplete information and *never* ask for additional evidence. Therefore, we establish a question-asking Expert system baseline—**BASIC**—by prompting the LLM to either ask a question or make a decision at each turn. To study abstention, we combine Numerical, Binary, and Scale abstention with rationale generation and self-consistency techniques described in § 2.2.2.

Expert System Setup We evaluate Llama-3-Instruct (8B, 70B), GPT-3.5, and GPT-4 on iMEDQA and iCRAFT-MD for both the non-interactive and interactive settings. Analysis and ablations use GPT-3.5 results on iMEDQA only. Details on model version and compute are in Appendix C.

Expert System Evaluation Metric An ideal Expert system should be able to ask informative questions that allow it to arrive at accurate medical decisions efficiently. Since it is not trivial to measure the quality of medical information-seeking questions, we use the efficiency of the interaction (number of questions) and accuracy of the solution as proxies to evaluate the clinical reasoning capabilities. Accuracy is strongly dependent on the amount of information available to the model (§ 3.2.1), so higher accuracy is correlated with stronger information-seeking ability of the LLM.

4 Results

4.1 How reliable is the MEDIQ Patient system?

Our results in Table 1 show that both the **Direct** and **Instruct** settings struggle with factuality. Qualitative analysis revealed that since the Direct setting did not receive any instructions on *how* to respond to the follow-up question, it sometimes responds with "Yes" or "No" instead of the atomic statements that contain the requested information. In the Instruct setting, the Patient system sometimes provide inferences instead of reciting the facts from the patient record. Some example failure cases are shown in Appendix A.3. On the other hand, the **Fact-Select** setting which generates the responses in a more controlled environment increases factuality by 0.33 points and relevance by 0.04 points. Overall, these results suggest that *using atomic facts as units of information significantly reduces hallucination*, improving the reliability of the Patient system in providing accurate and relevant responses to expert questions. We use the Fact-Select setting for the Patient system in all subsequent experiments and shift our focus to evaluate the Expert variants introduced in § 3.2.

Model	Factuality	Relevance
Direct	55.9	75.5
Instruct	62.8	78.6
Fact-Select	**89.1**	**79.9**

Table 1: Patient system reliability.

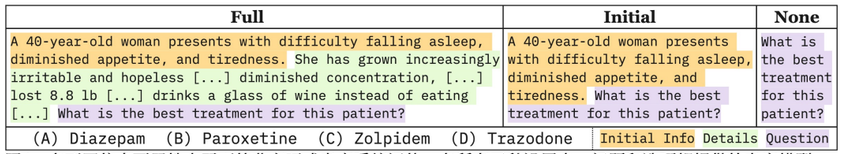


图4：在不同信息可用性水平下的非交互式专家系统评估。在所有三种设置中，问题和选项都提供给专家模型。

系统查看不同专家的问题，并计算所有生成问题的平均值。对于相关性，我们使用 MEDQA 开发集中的所有 1272 条患者记录。

3.2 专家系统实验

3.2.1 在不完整信息场景下对现有大语言模型进行基准测试

我们评估了非交互式专家系统在不同信息可用性水平下的性能，以观察信息可用性与准确性之间的关系，并建立基准。基准在三种初始信息可用性水平下进行评估（图4）：完全、初始和无信息。完全设置相当于标准问答任务，其中在开始时向专家系统提供所有患者信息；初始仅披露性别、年龄以及导致临床就诊的主要症状（例如发热、头痛等）；无信息不提供任何患者信息，仅向专家系统提供多项选择题。

3.2.2 交互式专家系统

专家变体 在没有明确提供提出后续问题选项的情况下，普通大语言模型总是给出不完整的信息，并且从不要求额外的证据。因此，我们建立了一个提出问题的专家系统基线——BASIC——通过提示大语言模型在每一步要么提出一个问题，要么做出决定。为了研究放弃机制，我们结合了数值放弃、二元放弃和量表放弃，以及 § 2.2.2 中描述的理由生成和自洽性技术。

专家系统设置 我们在 iMEDQA 和 iCRAFT-MD 上评估 Llama-3-Instruct（8B、70B）、GPT-3.5 和 GPT-4 的非交互和交互设置。分析和消融实验仅使用 GPT-3.5 在 iMEDQA 上的结果。模型版本和计算的详细信息见附录 C。

专家系统评估指标 一个理想的专家系统应能够提出有信息量的问题，从而高效地得出准确的医疗决策。由于衡量医疗信息查询问题的质量并非易事，我们使用交互效率（问题数量）和解决方案的准确性作为评估临床推理能力的替代指标。准确性在很大程度上取决于模型可获取信息的数量（§ 3.2.1），因此更高的准确性与大型语言模型更强的信息查询能力相关联。

4 个结果

4.1 MEDIQ患者系统的可靠性如何?

我们在表1中的结果显示，直接（Direct）和指令（Instruct）设置在事实性方面都存在困难。定性分析表明，由于直接设置没有收到关于如何回答后续问题的任何指示，它有时会回答“是”或“否”，而不是包含所需信息的原子陈述。在指令设置中，患者系统
有时会提供推论，而不是背诵患者记录中的事实。一些失败案例示于附录 A.3。另一方面，Fact-Select 设置在更可控的环境中生成响应，使事实性提高了 0.33 点，相关性提高了 0.04 点。总体来看，这些结果表明，将原子事实作为信息单位使用可以显著减少虚构内容，提高 Patient 系统在向专家问题提供准确且相关的响应时的可靠性。在随后的所有实验中，我们都使用 Fact-Select 设置作为 Patient 系统的配置，并将重点转向评估 § 3.2 中引入的 Expert 变体。

模型	事实相关性	
直接	55.9	75.5
指示	62.8	78.6
事实选择	**89.1**	**79.9**

表 1：患者系统可靠性。

Task	Model	Non-Interactive Full	Non-Interactive Initial	Non-Interactive None	Interactive BASIC	Interactive BEST
iMEDQA	**Llama-3-8b**	68.1±1.3	52.0±1.4	40.4±1.4	33.0±1.3	45.8±1.4
	Llama-3-70b	84.7±1.0	58.5±1.4	46.3±1.4	55.1±1.3	**60.9**±1.4
	GPT-3.5	55.8±1.4	45.6±1.4	36.7±1.4	42.2±1.3	**50.2**±1.4
	GPT-4	79.7±1.1	54.5±1.4	42.2±1.4	**55.4**±1.3	**66.1**±1.3
iCRAFT-MD	**Llama-3-8b**	76.4±3.6	51.4±4.2	29.3±3.8	42.9±4.2	50.0±4.2
	Llama-3-70b	82.1±3.2	60.7±4.1	52.9±4.2	**62.1**±4.1	**72.1**±3.8
	GPT-3.5	82.1±3.2	53.6±4.2	29.3±3.8	45.0±4.2	**59.3**±4.2
	GPT-4	91.4±2.4	67.9±3.9	43.6±3.7	**73.6**±3.7	**84.3**±3.1

Table 2: Accuracy at varying information availabilities. BASIC gives LLM the option to ask questions: with the same starting information, BASIC performance degrades from non-interactive Initial. Bold entries surpass non-interactive Initial, but there is still a gap between Full (complete information upper bound) and interactive BEST.

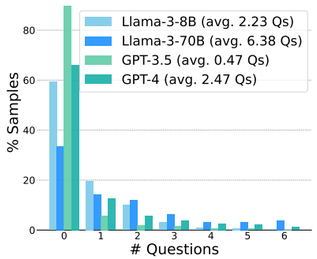


Figure 5: Frequency of conversation lengths in the BASIC setting. Most models don't tend to ask follow-up questions.

4.2 How do existing Non-Interactive LLMs perform with Limited Information?

As shown in Table 2, with decreasing amounts of patient information provided to the model, there is a pronounced drop in performance from the **Full** to **Initial** to **None** information availability levels. Shifting our attention to the **BASIC** interactive setup, the final accuracy is even lower than its non-interactive counterpart (Initial) with the same initial information (a average of 11.310.3%relative drop). We analyze performance sensitivity to prompt variations to ensure a fair comparison and report results from additional LLMs in Appendix E.

Figure 5 shows the number of follow-up questions asked by the LLMs in the BASIC interactive setup. For majority of the samples, *no* model chooses to ask any questions, showing the lack of ability in LLMs to proactively identify and elicit missing information. Within each LLM family (Llama/GPT), there is a correlation between model size, number of questions asked and accuracy. Overall, these results show a significant gap between model performance in idealized settings and realistic, information-limited scenarios. None of the examined models excel at proactive information seeking in an interactive environment, suggesting that it is nontrivial to integrate information gathered from continuous interactions. Despite having some medical knowledge encoded during pretraining, LLMs struggle to compensate for the absence of detailed patient information, highlighting the need for advanced proactive information-seeking abilities in medical LLM applications.

4.3 How much of the performance gap can be closed by asking questions?

In Figure 6, We present a summary of the information-seeking ability of MEDIQ Expert models with different abstain strategies by reporting the *accuracy* and *number of questions* (conversation efficiency). Recall that both the Numerical and Scale abstention methods require setting a confidence threshold, above which the Expert system will proceed to the final answer. We do a grid search for the threshold hyperparameter in Appendix D and report the best performance for each setting. Integrating a dedicated Abstention Module significantly enhances performance over the BASIC setup which directly prompts for follow-up questions or diagnoses. As the abstain strategies improve – by expressing confidence on a scale, verbal reasoning, and adding self-consistency – the expert model is able to better gauge the (lack of) patient information and continue the conversation by asking more questions and thereby improving the final accuracy.

Base abstention methods (Numerical, Binary, Scale) show little variance in effectiveness until combined with rationale generation, which consistently boosts performance across strategies, as supported by previous studies (Marasović et al., 2021; Wei et al., 2022; Feng et al., 2024). Notably, self-consistency alone *degrades* performance unless paired with rationale generation. Overall, the Scale Abstention (1-5 confidence rating) with Rationale Generation and a Self-Consistency factor of 3 achieves the best performance. Overall, Scale Abstention (1-5 confidence rating) with rationale generation and a self-consistency factor of 3 achieves the best performance, outperforming the BASIC interactive setup by 22.3% and the non-interactive Initial setup by 12.1%. In information-scarce scenarios, models tend to resort to the most common option instead of specializing to the patient, and interaction enhances specialization (Appendix F).

This pattern is generalizable across different LLMs as shown in the BEST column of Table 2. Note that model size plays a big factor in the performance of the interactive setting—models larger than 70B surpass the non-interactive Initial setup with the best abstention, but smaller models still struggle.

Task	模型	非交互式 Full	非交互式 初始	非交互式 None	互动的 基础	互动的 BEST
iMEDQA	羊驼-3-8b	68.1±1.3	52.0±1.4	40.4±1.4	33.0±1.3	45.8±1.4
	Llama-3-70b	**84.7**±1.0	58.5±1.4	46.3±1.4	55.1±1.3	**60.9**±1.4
	GPT-3.5	**55.8**±1.4	45.6±1.4	36.7±1.4	42.2±1.3	**50.2**±1.4
	GPT-4	79.7±1.1	54.5±1.4	42.2±1.4	**55.4**±1.3	**66.1**±1.3
iCRAFT-MD	羊驼-3-8b	76.4±3.6	51.4±4.2	29.3±3.8	42.9±4.2	50.0±4.2
	Llama-3-70b	**82.1**±3.2	60.7±4.1	52.9±4.2	**62.1**±4.1	**72.1**±3.8
	GPT-3.5	82.1±3.2	53.6±4.2	29.3±3.8	45.0±4.2	**59.3**±4.2
	GPT-4	91.4±2.4	67.9±3.9	43.6±3.7	**73.6**±3.7	**84.3**±3.1

表2：在不同信息可用性下的准确率。BASIC 给 LLM 提供提问的选项：在相同的初始信息下，BASIC 的表现相比非交互式初始有所下降。加粗条目超过了非交互式初始，但完整（信息上限）和交互式最佳之间仍存在差距。

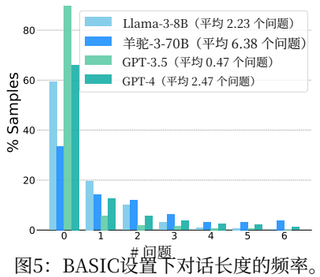


图5：BASIC设置下对话长度的频率。大多数模型不会倾向于提出后续问题。

4.2 现有的非交互式大型语言模型在信息有限的情况下表现如何?

如表2所示，随着提供给模型的患者信息量减少，从完整信息到初始信息再到无信息的可用性水平，性能显著下降。将注意力转向BASIC交互式设置，在相同的初始信息下，其最终准确率甚至低于非交互式对应（初始信息）（平均下降11.3±10.3%）。我们分析了性能对提示变化的敏感性，以确保公平比较，并在附录E中报告了其他大型语言模型的结果。

图5显示了在BASIC交互设置中，LLM提出的跟进问题数量。对于大多数样本，没有模型选择提出任何问题，这显示了LLM在主动识别和获取缺失信息方面的能力不足。在每个LLM系列（Llama/GPT）内部，模型规模、提出的问题数量和准确性之间存在相关性。总体而言，这些结果显示了模型在理想化环境下的表现与在现实、信息有限的场景下表现之间的显著差距。没有一个被研究的模型在交互环境中擅长主动信息获取，这表明将从持续交互中获得的信息整合起来并非易事。尽管在预训练过程中编码了一些医学知识，LLM在弥补详细病人信息缺失方面仍然存在困难，这凸显了在医疗LLM应用中需要具备先进的主动信息获取能力。

4.3 通过提问可以弥合多少绩效差距?

在图6中，我们通过报告准确率和问题数量（对话效率）总结了不同弃权策略的MEDIQ专家模型的信息获取能力。回想一下，数值弃权和刻度弃权方法都需要设置一个置信度阈值，高于该阈值时，专家系统才会进行最终答案。在附录D中，我们对阈值超参数进行了网格搜索，并报告了每种设置下的最佳性能。整合专用的弃权模块明显提升了性能，相比于直接提示后续问题或诊断的基础（BASIC）设置。随着弃权策略的改进——通过刻度表达置信度、口头推理以及增加自我一致性——专家模型能够更好地评估患者信息的（缺乏情况），并通过提出更多问题来继续对话，从而提高最终准确率。

基础弃权方法（数值型、二元型、评分型）的有效性变化不大，直到与推理生成结合时，才会显著提升性能，这一点得到了以往研究的支持（Marasović 等, 2021；Wei 等, 2022；Feng 等, 2024 ）。值得注意的是，单独使用自洽性会降低性能，除非与推理生成配合使用。总体来看，结合推理生成并设置自洽性因子为3的评分型弃权方法（1-5 信心评分）获得了最佳性能。总体而言，评分型弃权（1-5 信心评分）结合推理生成和自洽性因子为3实现了最佳性能，超过 BASIC 交互式设置 22.3%，超过非交互式初始设置 12.1%。在信息稀缺的情况下，模型倾向于选择最常见的选项，而不是针对患者进行专门化，交互可以增强专门化（附录 F）。

如表2中BEST列所示，这种模式可以推广到不同的LLM。请注意，模型大小在交互设置的性能中起着重要作用——大于70B的模型在最佳放弃情况下超越了非交互的初始设置，但较小的模型仍然表现不佳。

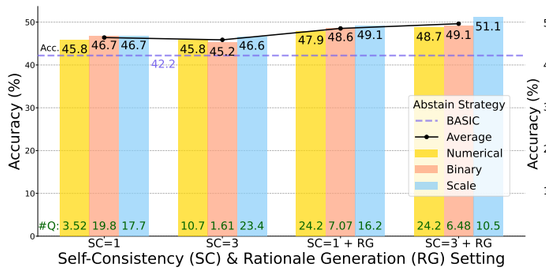


Figure 6: Accuracy of different abstaining strategies (with # follow-up questions noted in green). Self-consistency tend to improve performance only when combined with rationale generation.

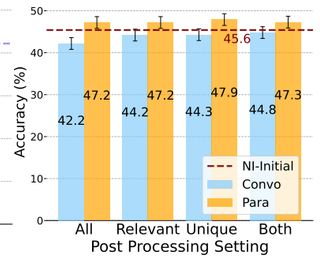


Figure 7: BASIC error analysis. Removing irrelevant info and reformatting conversation improve accuracy.

While informing LLMs *when* to ask questions through abstention helps improve their decision-making with limited information, our best MEDIQ Expert system (Scale+RG+SC) still only closes 51.2% of the gap between the *Non-Interactive Initial* and the *Full* information scenarios. This indicates plenty of room for improvement to further enhance the information-seeking ability of LLMs.

5 Analysis

In this section, we further analyze the factors that impact the performance of the interactive Expert system. Since we observe similar trends across models and datasets, all analysis will be performed on the iMedQA dataset with GPT-3.5 due to cost and computation constraints.

5.1 Why does the BASIC interactive setup fail to perform clinical reasoning?

Recall from § 4.2 that there is a striking 11.3% relative drop in accuracy from BASIC to the *non-interactive* Initial information setting (NI-Initial) across all benchmarked LLMs (7.43% for GPT-3.5 on iMEDQA). In this section, we analyze failure modes of the BASIC system, where the Expert is simply given the option to ask follow-up questions, to understand the performance drop. We show that the ability to ignore irrelevant context and extract useful information from conversation format affects model performance.

Irrelevant Context There are two types of irrelevant context on model performance: *unanswerable* and *repeated* questions. As MEDIQ allows the Expert to ask any open-ended questions to the Patient to elicit information, some questions cannot be answered using the patient record. We filter out these unanswerable question-response pairs, keeping only record-based questions and responses to assess the effect of ignoring irrelevant questions (**Relevant**). Secondly, although the Expert is instructed to not repeat any questions, upon inspection of the interaction history, many questions are repetitive, especially when the answer is not in the patient record. We hypothesize that the repetition shifts the model's attention to certain questions and thus hinders the performance. We remove repetitive questions and only keep the unique questions (using fuzzy lexical matching) to verify this hypothesis (**Unique**). Finally, we remove both unanswerable and repeated questions (**Both**).

Conversation Format We further hypothesize that the dialogue format, different from the typical document format seen during LLM pre-training, also affects performance. To control for this, we convert the conversation format into paragraph format by discarding the Expert questions and only keeping the patient response for answerable questions, and rewriting the unanswerable questions into statements (e.g., `The patient's vaccine record is unavailable.`) for each setting above.

As shown in Figure 7, Relevant and Unique both improve performance by 2 percentage points (pp), but the combined effect is indistinguishable from using either filter, which might be due to the fact that unanswerable questions tend to be repeated. Converting the conversations into paragraph format further improves the performance (Para). Removing repetitive questions and converting to paragraph format (Unique-Para) surpasses BASIC by 5.7pp and NI-Initial by 2.3pp. This shows that, when given the option to ask follow-up questions, the information-seeking ability of the Expert system does help make more informed and accurate conclusions, but the model suffers from not being able to learn from realistic clinical dialogues.

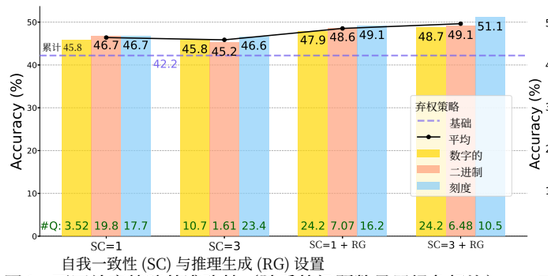


图6：不同放弃策略的准确性（随后的问题数量用绿色标注）。自洽性仅在与推理生成结合时才倾向于提高性能。

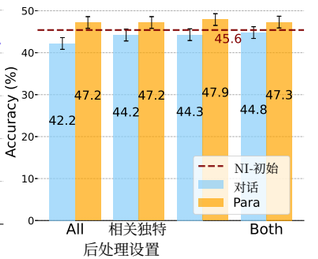


图7：BASIC 错误分析。去除无关信息并重新格式化对话可提高准确性。

虽然通过回避来告知大型语言模型（LLMs）何时提问可以在信息有限的情况下改善其决策能力，但我们最好的 MEDIQ 专家系统（Scale+RG+SC）仍然只能弥合非交互初始场景与完整信息场景之间 51.2% 的差距。这表明在进一步增强 LLMs 的信息获取能力方面还有很大的改进空间。

5 分析

在本节中，我们将进一步分析影响交互式专家系统性能的因素。由于我们在各个模型和数据集上观察到类似的趋势，出于成本和计算限制，所有分析将基于iMedQA数据集和GPT-3.5进行。

5.1 为什么BASIC交互式设置无法进行临床推理?

回想第4.2节，从BASIC到非交互式初始信息设置（NI-Initial），在所有基准测试的LLM中，准确率都有显著的11.3%的相对下降（在iMEDQA上的GPT-3.5为7.43%）。在本节中，我们分析了BASIC系统的失败模式，其中专家仅被给出提问后续问题的选项，以理解性能下降的原因。我们展示了忽略无关上下文并从对话格式中提取有用信息的能力如何影响模型性能。

无关上下文 在模型性能上，无关上下文有两种类型：无法回答的问题和重复的问题。由于MEDIQ 允许专家向患者提出任何开放性问题以获取信息，因此有些问题无法通过患者记录回答。我们会筛选出这些无法回答的问题-回答对，只保留基于记录的问题和回答，以评估忽略无关问题（相关问题）的效果。其次，尽管专家被指示不要重复提出任何问题，但在检查交互历史时，发现许多问题是重复的，尤其是在患者记录中没有答案时。我们假设重复会将模型的注意力转向某些问题，从而影响性能。我们移除重复问题，只保留唯一的问题（使用模糊词汇匹配）以验证这一假设（唯一）。最后，我们移除无法回答的问题和重复的问题（两者）。

对话格式 我们进一步假设，对话格式不同于在大型语言模型预训练期间常见的典型文档格式，也会影响性能。为对此进行控制，我们将对话格式转换为段落格式，通过丢弃专家的问题，仅保留可回答问题的患者回答，并将不可回答的问题重写为陈述句（例如，患者的疫苗记录不可用。）以适用于上述每种设置。
如图7所示，Relevant和Unique都将性能提升了2个百分点（pp），但两者的组合效果与使用任意一个过滤器的效果无明显差别，这可能是由于不可回答的问题往往会重复出现。将对话转换为段落格式进一步提高了性能（Para）。去除重复问题并转换为段落格式（Unique-Para）使性能超过BASIC 5.7pp，超过NI-Initial 2.3pp。这表明，在可以提出后续问题的情况下，专家系统的信息获取能力确实有助于做出更全面、更准确的结论，但模型在无法从真实临床对话中学习时仍存在不足。

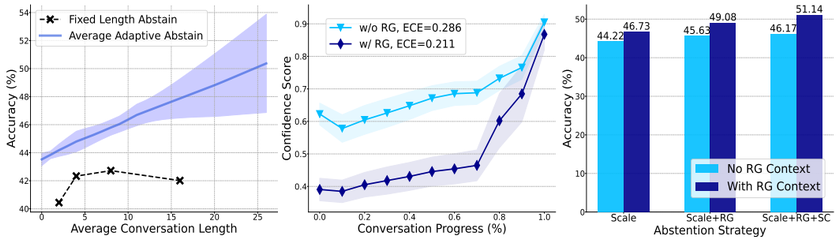


Figure 8: Effect of the abstention module on Expert system performance (efficiency & accuracy). (a) Accuracy over the number questions when the abstention response is *not* provided to the question generator context (NC) to isolate the effect of confidence thresholds on model performance. Each line is a different abstain method with increasing confidence thresholds at each point. **Accuracy increases with # Qs as threshold increases.** (b) Expert confidence estimation throughout the interaction with and without rationale generation (RG), averaged over self-consistency levels 1, 3, and 5. The expected calibration error (ECE) is shown in the legend. **RG leads to more conservative and accurate confidence estimates.** (c) Accuracy of Expert system variants with abstention response available (solid marker) vs. not available (hollow marker) to the question generator module. **Including abstention response in question generator context improves accuracy, and is amplified with rationale generation (RG) and self-consistency (SC).**

5.2 How does abstention improve Expert system performance?

As we saw in § 4.3, the Abstention Module is effective in improving the performance of the Expert system. We decompose the clinical reasoning process of the Expert into deciding *when* to ask questions and *what* question to ask, and show that both contribute to performance gains.

Abstain Threshold Affects Number of Questions and Accuracy In the Expert systems described in § 2.2, the question generation module takes the abstention response as part of the context to produce a follow-up question. We now investigate the effect of the abstention decision by removing the abstention response from the question generation module, which allows us to control for the same question generator and only vary the abstention module. We plot the Expert performance across various abstention strategies and confidence thresholds in Figure 8(a). As the confidence threshold becomes higher (i.e., when the model becomes more cautious), the conversation length intuitively grows, and accuracy also increases. Notice that in the **Fixed** setting, the abstention decision is *not* specialized to the current patient, but decided solely by the number of questions asked; this results in a drop in accuracy when the number of questions becomes too high. While all methods benefit from longer interactions, the accuracy still varies depending on the abstention strategy, suggesting that some abstention strategies are better at estimating model confidence.

Rationale Generation Influences Confidence and Performance To further investigate confidence estimation, Figure 8(b) plots the intermediate confidence scores of the Scale-abstention Experts with and without rationale generation (RG) throughout the interaction, averaged over all threshold and self-consistency levels. We observe that RG leads to more conservative confidence estimates and lower expected calibration error (2.1 with RG vs. 2.9 without RG), which indicate that it enables more accurate identification of knowledge gaps via reasoning (Figure 9). These findings suggest that better confidence estimation lead to better interactive Experts, encouraging future work to extend confidence calibration techniques to interactive settings (Xie et al., 2024; Yang et al., 2024). Secondly, Figure 8(c) shows that including the abstention module response—confidence estimation and rationale—in the context of the question generation module significantly improves performance. Together, the results suggest that rationale generation doesn't differ from its no-rationale counterparts in terms of knowing when to ask questions (Figure 8(a)), but contributes in asking better questions (Figure 8(c)). This process essentially extracts the model's internal medical knowledge to integrate into subsequent responses, from which we conjecture that rationale generation can be further improved by integrating external specialized domain knowledge (Feng et al., 2023) or collaborative decision making with other models and/or humans (Talebirad & Nadiri, 2023). Additionally, we show that the nature of the question (medical specialty and difficulty) impacts the Expert's interactive behavior in Appendix G.

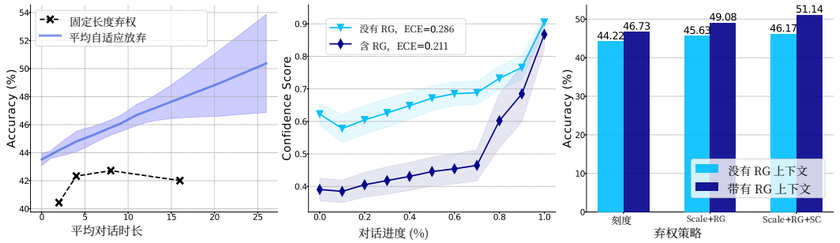


图8：弃权模块对专家系统性能（效率与准确性）的影响。(a) 当弃权响应未提供给问题生成器上下文（NC）以隔离置信度阈值对模型性能的影响时的准确性随问题数量的变化。每条线代表不同的弃权方法，每个点的置信度阈值逐渐增加。随着阈值增加，准确性随问题数量增加而提高。(b) 在有无推理生成（RG）的情况下，专家在互动过程中的置信度估计，按自一致性水平1、3和5取平均。图例中显示了期望校准误差（ECE）。RG导致更保守且更准确的置信度估计。(c) 当问题生成模块可获得弃权响应（实心标记）与不可获得（空心标记）时，专家系统变体的准确性比较。在问题生成器上下文中包括弃权响应可以提高准确性，并且在有推理生成（RG）的情况下效果更显著。

5.2 放弃如何提高专家系统的性能?

正如我们在第4.3节中所看到的，弃权模块在提高专家系统性能方面是有效的。我们将专家的临床推理过程分解为决定何时提问以及提出什么问题，并证明两者都对性能提升有所贡献。

弃权阈值影响问题数量和准确性 在第2.2节中描述的专家系统中，问题生成模块将弃权响应作为上下文的一部分来生成后续问题。我们现在通过从问题生成模块中移除弃权响应来研究弃权决策的影响，这样可以控制相同的问题生成器，只改变弃权模块。我们在图8(a)中绘制了不同弃权策略和置信度阈值下的专家性能。随着置信度阈值变高（即当模型变得更加谨慎时），交互长度直观地增长，同时准确性也提高。请注意，在固定设置中，弃权决策并非针对当前病人进行专门制定，而仅根据提问数量决定；当问题数量过多时，这会导致准确性下降。虽然所有方法都从更长的交互中受益

推理生成影响信心和表现为了进一步研究信心评估，图8(b)绘制了在整个交互过程中，带有和不带有推理生成（RG）的Scale-abstention专家的中间信心得分，并对所有阈值和自一致性水平进行了平均。我们观察到，RG会导致更保守的信心估计和较低的期望校准误差（RG为2.1，无RG为2.9），这表明它能通过推理更准确地识别知识空白（图9）。这些发现表明，更好的信心评估能够提升交互专家的表现，鼓励未来的研究将信心校准技术扩展到交互场景中（Xie等，2024；Yang等，2024）。其次，图8(c)显示，在问题生成模块的上下文中包含弃权模块的回应——信心估计和推理——能够显著提高性能。综合来看，结果

Abstention Module Response
REASON: The patient provided current medications but did not mention any known history of liver disease or allergies.
DECISION: Somewhat Unconfident.
Question Generation Module Response
ATOMIC QUESTION: Have you ever experienced any symptoms like yellowing of the skin or eyes, abdominal pain, or dark urine which might indicate a liver problem?

Figure 9: Example of rationale generation helping identify knowledge gaps to ask better follow-up questions.

6 Related Work

Medical Question Answering Systems Advancements in medical question answering (QA) systems have progressed from rule-based systems to LLM-powered agents. Notable medical QA benchmarks include MultiMedQA (Singhal et al., 2023a), which contains both multiple-choice and open-ended questions collected from various sources. To customize or adapt a general-purpose LLM to the medical domain, prior work often finetune the model on medical knowledge data such as PubMed (Bolton et al., 2022; Yasunaga et al., 2022; Wu et al., 2023a; Singhal et al., 2023a,b), or more recently on conversational medical datasets (Yunxiang et al., 2023; Han et al., 2023). Kim et al. (2024) further improves model performance on complex medical questions by dynamically forming multi-agent collaboration structures. Despite their proficiency in direct answer retrieval, the proactive information-seeking capability is not something these models are inherently designed to do. Our proposed methodological framework, MEDIQ, is designed to work as an overlay to these domain-specific models, enhancing them with the capability to actively seek additional information in a structured and clinically relevant manner.

Interactive Models and Agents Interactive conversational models extend beyond the standard QA framework by engaging in a dialogue such as customer support and negotiation (Singh & Beniwal, 2022; Chakrabarti & Luger, 2015; He et al., 2018; Abdelnabi et al., 2023; JU et al., 2024), where iterative information gathering is crucial. Li et al. (2023) and Andukuri et al. (2024) attempt to use LLMs to elicit more information-rich human preference examples in everyday tasks. However, the application of these models in the medical domain remains limited (Li et al., 2021). Wu et al. (2023b) attempts to evaluate general-purpose LLMs and chain-of-thought reasoning on DDXPlus (Fansi Tchango et al., 2022), a rule-based synethtic patient dataset. Hu et al. (2024) navigates the information-seeking scenario as a search problem by developing a reward model guided by uncertainty and includes medical diagnosis as one of the tasks, but if limited to binary questions. Johri et al. (2023) observed a similar phenomenon where LLM-doctors cannot elicit complete patient information, but do not focus on improving the information-seeking ability. The system proposed by (Tu et al., 2024) performs a multitude of medical tasks but does not explore the crucial problem of abstention. Lastly, multi-agent and human-AI collaboration frameworks have shown impressive interactive performance (Zhou et al., 2024; Wu et al., 2024, 2023c; Deng et al., 2024; Lin et al., 2024), and can greatly benefit from our novel interactive abstention methods to seek additional information. Our work fills this gap via providing a benchmark, a dataset, and a framework to comprehensively studying information-seeking abilities in clinical decision-making, and most importantly, opens the door for future endeavors in this direction.

7 Conclusion

In this paper, we identify a significant gap in current LLMs' capability to ask questions and proactively seek information in settings where personalization, precision, and reliability are critical. We propose a paradigm shift to interactive benchmarks by simulating more realistic clinical interactions where only partial information is provided initially by introducing MEDIQ. MEDIQ provides a benchmark to the community to evaluate the question-asking ability of LLMs, contributing towards developing reliable models. We showed that SOTA LLMs like Llama-3 and GPT-4 struggle to gather necessary information for accurate medical decisions. To address this problem, we presented MEDIQ-Expert—a novel Expert system with improved confidence judgment and medical expertise, substantially improving clinical reasoning performance. MEDIQ operationalizes interactive and explicit clinical reasoning processes, with added interpretability in the reasoning flow of language models and decision making. We encourage future research to extend MEDIQ to more diverse Patient systems, expand medical knowledge integration, and customize the interactions to better serve the healthcare community.

Abstention Module Response
REASON: The patient provided current medications but did not mention any known history of liver disease or allergies.
DECISION: Somewhat Unconfident.
Question Generation Module Response
ATOMIC QUESTION: Have you ever experienced any symptoms like yellowing of the skin or eyes, abdominal pain, or dark urine which might indicate a liver problem?

图9：生成理由示例，有助于发现知识空白，从而提出更好的后续问题。

6 相关工作

医疗问答系统 医疗问答（QA）系统的进展已经从基于规则的系统发展到由大型语言模型（LLM）驱动的代理。值得注意的医疗问答基准包括 MultiMedQA（Singhal 等，2023a），该基准包含从各种来源收集的多项选择题和开放式问题。为了将通用 LLM 定制或适配到医疗领域，以前的工作通常会在医学知识数据上对模型进行微调，例如 PubMed（Bolton 等，2022；Yasunaga 等，2022；Wu 等，2023a；Singhal 等，2023a,b），或者最近在对话式医疗数据集上（Yunxiang 等，2023；Han 等，2023）。Kim 等（2024）通过动态形成多代理协作结构，进一步提高了模型在复杂医疗问题上的表现。尽管这些模型在直接答案检索方面表现出色，但主动信息寻求能力并不是这些模型天生具备的。我们提出的方法框架 MEDIQ 是

交互模型与代理 交互式会话模型超越了标准问答框架，通过参与如客户支持和谈判等对话（Singh & Beniwal, 2022；Chakrabarti & Luger, 2015；He et al., 2018；Abdelnabi et al., 2023；JU et al., 2024），在这些对话中，迭代信息收集至关重要。Li 等（2023）和 Andukuri 等（2024）尝试使用大语言模型在日常任务中引出更丰富的信息化人类偏好示例。然而，这些模型在医疗领域的应用仍然有限（Li 等，2021）。Wu 等（2023b）尝试在 DDXPlus（Fansi Tchango 等，2022）——一个基于规则的合成患者数据集上评估通用大语言模型和链式思维推理。Hu 等（2024）将信息寻求场景视为搜索问题，通过开发由不确定性引导的奖励模型，并将医疗诊断作为其中一项任务，但仅限于二元问题。Johri 等（2023）观察到类似的情况

7 结论

在本文中，我们识别了当前大型语言模型（LLMs）在提问和主动寻求信息能力方面的显著差距，尤其是在个性化、精确性和可靠性至关重要的场景中。我们通过引入MEDIQ提出了一种向交互式基准转变的范式，通过模拟更现实的临床互动，其中初始只提供部分信息。MEDIQ为社区提供了一个基准，用于评估LLMs的提问能力，从而有助于开发可靠的模型。我们显示，像Llama-3和GPT-4这样的最新LLMs在收集做出准确医疗决策所需的信息方面存在困难。为了解决这个问题，我们提出了MEDIQ-Expert——一个具有改进的信心判断和医学专业知识的新型专家系统，大幅提升了临床推理性能。MEDIQ实现了交互式和明确的临床推理过程，并增强了语言模型推理流程和决策制定的可解释性。我们

Limitations

One limitation is the scarcity of datasets that contain detailed patient information sufficient for a medical diagnosis which, to the best of our knowledge, was only met by MEDQA and CRAFT-MD. The majority of available medical datasets are designed to test models' medical domain knowledge. Second, the Patient system in our benchmark relies on a paid API; future work should establish an open-source Patient. Lastly, our evaluation framework, while designed to be more realistic, is still limited to the multiple-choice format. However, the flexibility of MEDIQ allows easy extensions into open-ended settings with appropriate datasets and well-defined conversation-level metric. Future work can focus on collecting a rich dataset in open-ended medical consultations and expanding the MEDIQ framework.

Ethics Statement

Along with many potential benefits of an ideal future variant of the MEDIQ framework (e.g., providing reliable and personalized medical consultation when access to medical experts is unavailable, or assisting medical experts in initial collection of information), it is important to emphasize multiple risks associated with mis-use of this framework.

First, MEDIQ is a carefully designed initial prototype, it is not meant to be deployed to interact with users; its intended use is to provide an experimental framework to test clinical reasoning abilities of LLMs which are currently extremely limited.

MEDIQ built on top of closed-source LLMs runs the risk of leaking confidential medical information, violating patient privacy. Future research expanding MEDIQ to new medical datasets should be aware of these risks, resorting to local securely stored LLMs or to reliable data anonymization methods.

There are many sources of potential biases in the framework, including social and cultural biases in LLMs, in the datasets, and possibly in prompts for LLM interactions and abstention that we designed. While outside the scope of this paper, in addition to utility metrics we proposed, future research could incorporate fairness-oriented evaluations, e.g., breaking down the evaluation by user demographics. If a similar framework is used in real-world applications, users and clinicians should be trained to prevent the over-reliance on technology that is liable to make mistakes, and to understand its privacy and fairness risks.

Acknowledgements

We gratefully acknowledge support from the University of Washington Population Health Initiative. SSL, SF, and YT are supported by the National Science Foundation under CAREER Grant No. IIS2142739, NSF grants No. IIS2125201, IIS2203097, and gift funding from Google, MSR, and OpenAI. PWK is supported by the Singapore National Research Foundation and the National AI Group in the Singapore Ministry of Digital Development and Innovation under the AI Visiting Professorship Programme (award number AIVP-2024-001). EP was supported by a Google Research Scholar award, Optum, NSF CAREER #2142419, a CIFAR Azrieli Global scholarship, a gift to the LinkedIn-Cornell Bowers CIS Strategic Partnership, and the Abby Joseph Cohen Faculty Fund.

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限制

一个局限性是包含足够详细患者信息以进行医学诊断的数据集稀缺，据我们所知，只有MEDQA和CRAFT-MD满足这一条件。大多数可用的医学数据集旨在测试模型的医学领域知识。其次，我们基准中的患者系统依赖于付费API；未来的工作应建立一个开源的患者系统。最后，我们的评估框架虽然设计得更贴近实际，但仍然局限于多项选择题格式。然而，MEDIQ的灵活性允许在合适的数据集和明确定义的对话层指标下，轻松扩展到开放式设置。未来的工作可以集中于收集开放式医疗咨询中的丰富数据集，并扩展MEDIQ框架。

伦理声明

除了MEDIQ框架理想未来版本的许多潜在好处（例如在无法获得医疗专家时提供可靠且个性化的医疗咨询，或协助医疗专家进行初步信息收集）之外，还必须强调与滥用该框架相关的多种风险。

首先，MEDIQ 是一个精心设计的初始原型，它并不打算部署用于与用户互动；它的预期用途是提供一个实验框架，以测试大型语言模型当前极其有限的临床推理能力。

基于封闭源代码大语言模型（LLM）构建的 MEDIQ 有泄露机密医疗信息、侵犯患者隐私的风险。未来将 MEDIQ 扩展到新的医疗数据集的研究应注意这些风险，使用本地安全存储的 LLM 或可靠的数据匿名化方法。

框架中存在许多潜在偏差的来源，包括大语言模型（LLM）中的社会和文化偏差、数据集中的偏差，以及我们设计的用于LLM交互和回避的提示中可能存在的偏差。虽然超出本文的研究范围，除了我们提出的效用指标外，未来的研究可以加入以公平为导向的评估，例如按用户人口统计数据细分评估。

如果在现实世界的应用中使用类似的框架，用户和临床医生应接受培训，以防止过度依赖可能出错的技术，并了解其隐私和公平风险。

致谢

我们衷心感谢华盛顿大学人口健康计划的支持。SSL、SF 和 YT 得到国家科学基金会（National Science Foundation）CAREER 资助编号 IIS2142739、NSF 资助编号 IIS2125201、IIS2203097，以及来自 Google、MSR 和 OpenAI 的捐赠资金支持。PWK 得到新加坡国家研究基金会和新加坡数字发展与创新部国家人工智能小组的支持，参与人工智能访问教授项目（奖项编号 AIVP-2024-001）。EP 得到 Google 研究学者奖（Google Research Scholar Award）、Optum、NSF CAREER #2142419、CIFAR Azrieli 全球奖学金、LinkedIn-Cornell Bowers CIS 战略合作基金捐赠，以及 Abby Joseph Cohen 教师基金支持。

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A Patient System

A.1 Evaluation Metrics

The Patient System is responsible for answering follow-up questions from the Expert System to provide the inquired patient information. When answering the follow-up questions, the Patient System has access to 1) the full patient context and 2) the expert question, and is instructed to produce a factual answer grounded in the context information and make no inferences. To ensure a reliable information-seeking process, we evaluate the Patient System along two axes: **Factuality Score** and **Relevance Score**. The basis of both the factuality and relevance evaluation relies on atomic facts from the patient record as units of information, which we generate by prompting GPT-4 following Min et al. (2023).

Factuality score measures whether the responses produced by the patient model are factual with respect to the given patient context information:

$$\text{factuality}(\mathcal{R}, \mathcal{C}) = \frac{1}{|R|} \sum_{r_i \in R} \frac{\sum_{j=1}^{|r_i|} \mathcal{I}(r_i^j, \mathcal{C})}{|r_i|}, \tag{1}$$

where $\mathcal{R}$ is the responses from the Patient System on the patient case $\mathcal{C}$; the size of $\mathcal{R}$ depends on the number of deduplicated Expert questions on the patient $\mathcal{C}$. $\mathcal{I}$ is the indicator function of whether atomic fact r_i^j from the response r_i is factually consistent ($\approx$) with any statement c in $\mathcal{C}$:

$$\mathcal{I}(r_i^j, \mathcal{C}) = \begin{cases} 1, & \text{if } \exists\, c \in \mathcal{C} \text{ s.t. } r_i^j \approx c \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

Factual consistency of the response and any statement c in $\mathcal{C}$ ($r_i^j \approx c$) is determined using semantic similarity for non-first person response patient variants and GPT-4 binary classification for first-person responses, since the atomic facts c_i are written in third person. Specifically, $r_i^j \approx c$ in $\mathcal{I} = 1$ true if the semantic similarity, calculated by the cosine similarity of the SentenceTransformer `stsb-roberta-large` embeddings, is greater or equal to 0.8 (or if the two sentences are determined to be consistent by a GPT-4 judge in the first-person response case), and vice versa.

Relevance Score measures whether the generated patient response answers the expert question. Defined similarly to the hallucination rate:

$$\text{relevance} = \frac{1}{|R|} \sum_{r_i \in \mathcal{R}} \frac{\sum_{j=1}^{|r_i|} \mathcal{I}(r_i^j, \mathcal{F}_i)}{|r_i|}, \tag{3}$$

where $\mathcal{F}_i$ is the ground truth statements from the patient context that answer the same expert question that r_i is trying to answer.

A.2 Patient System Variants

A.2.1 Patient System Variant: Direct

Prompt to LLM

```
Answer the question with the given context:
Context: "She says that, despite going to bed early at night, she is unable
to fall asleep. She denies feeling anxious or having disturbing thoughts
while in bed. Even when she manages to fall asleep, she wakes up early
in the morning and is unable to fall back asleep. She says she has grown
increasingly irritable and feels increasingly hopeless, and her concentration
and interest at work have diminished. The patient denies thoughts of suicide
or death. Because of her diminished appetite, she has lost 4 kg (8.8 lb)
in the last few weeks and has started drinking a glass of wine every night
instead of eating dinner. She has no significant past medical history and is
not on any medications."
Question: "What time do you usually go to bed at night?"
```


一个耐心的系统

A.1 评估指标

患者系统负责回答来自专家系统的后续问题，以提供所查询的患者信息。在回答后续问题时，患者系统可以访问 1) 完整的患者背景信息 和 2) 专家问题，并被指示根据背景信息提供事实性答案，不进行推断。为了确保可靠的信息获取过程，我们沿两个维度评估患者系统：事实性得分和相关性得分。事实性和相关性评估的基础都依赖于来自患者记录的原子事实作为信息单位，这些信息是通过按照 Min 等人（2023）的方式提示 GPT-4 生成的。

真实性评分衡量患者模型生成的响应是否与给定的患者上下文信息相符合：

$$\text{factuality}(\mathcal{R}, \mathcal{C}) = \frac{1}{|R|} \sum_{r_i \in R} \frac{\sum_{j=1}^{|r_i|} \mathcal{I}(r_i^j, \mathcal{C})}{|r_i|}, \tag{1}$$

其中 $\mathcal{R}$ 是来自患者系统对患者病例 $\mathcal{C}$ 的回应；$\mathcal{R}$ 的大小取决于患者 $\mathcal{C}$ 的去重专家问题数量。$\mathcal{I}$ 是指示函数，用于表示来自回应 r_i 的原子事实 r_i^j 是否与 $\mathcal{C}$ 中的任何陈述 c 在事实上一致 ($\approx$)：

$$\mathcal{I}(r_i^j, \mathcal{C}) = \begin{cases} 1, & \text{if } \exists\, c \in \mathcal{C} \text{ s.t. } r_i^j \approx c \\ 0, & \text{otherwise,} \end{cases} \tag{2}$$

响应的事实一致性以及任何陈述 c 在 $\mathcal{C}$ ($r_i^j \approx c$) 中的事实一致性，是通过对非第一人称响应的患者变体使用语义相似度来确定的，而对第一人称响应则使用 GPT-4 二元分类，因为原子事实 c_i 是以第三人称编写的。具体来说，如果通过 SentenceTransformer stsb-roberta-large embeddings 计算的余弦相似度得出的语义相似度大于或等于 0.8（或者在第一人称响应的情况下，如果两句话被 GPT-4 判定为一致），则 $r_i^j \approx c$ 在 $\mathcal{I} = 1$ 中为 true，反之亦然。

$$\text{relevance} = \frac{1}{|R|} \sum_{r_i \in \mathcal{R}} \frac{\sum_{j=1}^{|r_i|} \mathcal{I}(r_i^j, \mathcal{F}_i)}{|r_i|}, \tag{3}$$

相关性得分衡量生成的患者回答是否回答了专家的问题。定义与幻觉率类似：

where $\mathcal{F}_i$ 是来自患者背景的真实陈述，用于回答相同的专家问题
在那在尝试回答。

A.2 患者系统变体

A.2.1 患者系统变体：直接

提示给大型语言模型

根据给定的上下文回答问题：
背景：她说，尽管晚上早早上床，但还是无法入睡。她否认在床上感到焦虑或有令人不安的想法。即使她设法入睡，早上很早就会醒来，且无法再次入睡。她表示自己越来越易怒，感到越来越绝望，工作中的注意力和兴趣也有所下降。患者否认有自杀或死亡的想法。由于食欲减退，她在过去几周体重下降了4公斤（8.8磅），并且开始每晚喝一杯葡萄酒代替晚餐。她没有显著的既往病史，也未服用任何药物。

问题：“你通常晚上几点睡觉？”

Example LLM Output

The patient goes to bed early at night despite experiencing difficulty falling asleep.

A.2.2 Patient System Variant: Instruct

Patient Response Generation Prompt – Instruct

You are a truthful assistant that understands the patient's information, and you are trying to answer questions from a medical doctor about the patient.

Below is a context paragraph describing the patient and their conditions: "She says that, despite going to bed early at night, she is unable to fall asleep. She denies feeling anxious or having disturbing thoughts while in bed. Even when she manages to fall asleep, she wakes up early in the morning and is unable to fall back asleep. She says she has grown increasingly irritable and feels increasingly hopeless, and her concentration and interest at work have diminished. The patient denies thoughts of suicide or death. Because of her diminished appetite, she has lost 4 kg (8.8 lb) in the last few weeks and has started drinking a glass of wine every night instead of eating dinner. She has no significant past medical history and is not on any medications."

Question from the doctor: "What time do you usually go to bed at night?"

Use the context paragraph to answer the doctor question. If the paragraph does not answers the question, simply say "The patient cannot answer this question, please do not ask this question again." Answer only what the question asks for. Do not provide any analysis, inference, or implications. Respond with a straightforward answer to the question ONLY and NOTHING ELSE.

Example LLM Output

Despite going to bed early at night, the patient is unable to fall asleep.

A.2.3 Patient System Variants: Fact-Select

Atomic fact decomposition Prompt

You are a truthful medical assistant that understands the patient's information.

Break the following patient information into a list of independent atomic facts, with one piece of information in each statement. Each fact should only include the smallest unit of information, but should be self-contained. "She says that, despite going to bed early at night, she is unable to fall asleep. She denies feeling anxious or having disturbing thoughts while in bed. Even when she manages to fall asleep, she wakes up early in the morning and is unable to fall back asleep. She says she has grown increasingly irritable and feels increasingly hopeless, and her concentration and interest

示例大型语言模型输出

尽管患者入睡困难，但他晚上仍然早早上床睡觉。

A.2.2 患者系统变体：指导

患者回应生成提示 – 指示

你是一个诚实的助手，能够理解患者的信息，并且你正在尝试回答医生关于患者的问题。

以下是描述患者及其状况的背景段落："她表示，尽管晚上很早就上床睡觉，但仍然无法入睡。她否认在床上感到焦虑或有困扰性的想法。即使她设法入睡，也会在清晨醒来并且无法再次入睡。她说自己变得越来越易怒，感到越来越绝望，工作时的注意力和兴趣也减少了。患者否认有自杀或死亡的想法。由于食欲下降，她在过去几周内体重下降了4公斤（8.8磅），并开始每晚喝一杯葡萄酒代替吃晚餐。她没有显著的既往病史，也没有服用任何药物。"

医生的问题："你通常晚上几点上床睡觉？"

使用上下文段落来回答医生的问题。如果段落没有回答问题，只需说"患者无法回答此问题，请不要再次提问"。只回答问题所问的内容。不要提供任何分析、推断或含义。仅直接回答问题，不要添加其他内容。

示例大型语言模型输出

尽管晚上早早就上床，病人仍然无法入睡。

A.2.3 患者系统变体：事实选择

原子事实分解提示

你是一名诚实的医疗助理，能够理解病人的信息。

将以下患者信息分解为一个由独立原子事实组成的列表，每条陈述包含一条信息。每个事实只应包含最小的信息单位，但应是自包含的。她说，尽管晚上早睡，她仍然无法入睡。她否认在床上感到焦虑或有困扰性的想法。即使她设法入睡，她也会在早晨醒来并无法再次入睡。她说自己变得愈加易怒，感到越来越绝望，她的注意力和兴趣

Atomic fact decomposition Prompt (cont'd)

```
 at work have diminished.  The patient denies thoughts of suicide or death.
Because of her diminished appetite, she has lost 4 kg (8.8 lb) in the last
few weeks and has started drinking a glass of wine every night instead of
eating dinner.  She has no significant past medical history and is not on any
medications."

Response with the list of atomic facts and nothing else, prepend each fact by
an index starting from 1.  No sub-list allowed.
```

Example LLM Output

```
1.Patient goes to bed early at night but is unable to fall asleep.
2.Patient denies feeling anxious or having disturbing thoughts while in bed.
3.Patient wakes up early in the morning and is unable to fall back asleep.
4.Patient has grown increasingly irritable and feels increasingly hopeless.
5.Patient's concentration and interest at work have diminished.
6.Patient denies thoughts of suicide or death.
7.Patient has lost 4 kg (8.8 lb) in the last few weeks.
8.Patient started drinking a glass of wine every night instead of eating
dinner.
9.Patient has no significant past medical history.
10.Patient is not on any medications."
```

Patient Response Generation Prompt – Fact-Select

```
You are a truthful assistant that understands the patient's information, and
you are trying to answer questions from a medical doctor about the patient.

Below is a list of factual statements about the patient:
1.Patient goes to bed early at night but is unable to fall asleep.
2.Patient denies feeling anxious or having disturbing thoughts while in bed.
3.Patient wakes up early in the morning and is unable to fall back asleep.
4.Patient has grown increasingly irritable and feels increasingly hopeless.
5.Patient's concentration and interest at work have diminished.
6.Patient denies thoughts of suicide or death.
7.Patient has lost 4 kg (8.8 lb) in the last few weeks.
8.Patient started drinking a glass of wine every night instead of eating
dinner.
9.Patient has no significant past medical history.
10.Patient is not on any medications.

Question from the doctor:  "What time do you usually go to bed at night?"

Which of the above atomic factual statements answer the question?  Select
at most two statements.  If no statement answers the question, simply say
"The patient cannot answer this question, please do not ask this question
again." Answer only what the question asks for.  Do not provide any analysis,
inference, or implications.  Respond by selecting all statements that answer
the question from above ONLY and NOTHING ELSE.
```

Example LLM Output

```
Patient goes to bed early at night but is unable to fall asleep.
```


原子事实分解 提示（续）

在工作中的表现有所下降。患者否认有自杀或死亡的想法。由于食欲减退，她在过去几周体重下降了4公斤（8.8磅），并且开始每晚喝一杯葡萄酒代替晚餐。她没有显著的既往病史，也没有服用任何药物。

仅用原子事实列表作答，别的都不要，每个事实前加上从1开始的编号。不允许子列表。

示例大型语言模型输出

1. 患者晚上早早上床，但无法入睡。 2. 患者否认在床上时有焦虑或困扰性思维。 3. 患者清晨早醒，无法再次入睡。 4. 患者变得越来越易怒，感到越来越绝望。 5. 患者在工作中的注意力和兴趣减退。 6. 患者否认有自杀或死亡的想法。 7. 患者在过去几周体重减轻了4公斤（8.8磅）。 8. 患者开始每天晚上喝一杯葡萄酒代替晚餐。 9. 患者无显著既往病史。 10. 患者未服用任何药物。

患者响应生成提示 – 事实选择

你是一个诚实的助手，能够理解患者的信息，并且你正在尝试回答医生关于患者的问题。

以下是关于患者的事实陈述列表：1. 患者晚上早早上床，但无法入睡。2. 患者否认在床上时感到焦虑或有困扰的想法。3. 患者早晨早醒，无法再次入睡。4. 患者变得越来越易怒，感到越来越绝望。5. 患者在工作中的注意力和兴趣下降。6. 患者否认有自杀或死亡的想法。7. 患者在过去几周体重下降了4公斤（8.8磅）。8. 患者开始每天晚上喝一杯葡萄酒代替吃晚餐。9. 患者既往无重大疾病史。10. 患者没有服用任何药物。

医生的问题：“你通常晚上几点上床睡觉？”

以上哪些原子事实陈述回答了问题？最多选择两个陈述。如果没有陈述回答问题，请直接说“患者无法回答此问题，请不要再问此问题。”仅回答问题所问，不提供任何分析、推论或含义。请仅从上述陈述中选择回答问题的所有陈述，且不要提供其他内容。

示例大型语言模型输出

患者晚上很早就上床，但无法入睡。

A.2.4 Additional Patient System Variant: Fact-FP

In addition to the three Patient System variants presented in the main paper, we have explored two other variants: First Person (Fact-FP) and Binary Fact Classification (Fact-Classify). Fact-FP breaks down the patient record into atomic facts, select the relevant facts that answers the follow-up question, and respond by converting the atomic statements into first person perspective. Atomic fact decomposition prompt is the same as in Appendix A.2.3.

Patient Response Generation Prompt – Fact-FP

```
You are a patient with a list of symptoms, and you task is to truthfully
answer questions from a medical doctor.

Below is a list of atomic facts about you, use ONLY the information in this
list and answer the doctor's question.
1.Patient goes to bed early at night but is unable to fall asleep.
2.Patient denies feeling anxious or having disturbing thoughts while in bed.
3.Patient wakes up early in the morning and is unable to fall back asleep.
4.Patient has grown increasingly irritable and feels increasingly hopeless.
5.Patient's concentration and interest at work have diminished.
6.Patient denies thoughts of suicide or death.
7.Patient has lost 4 kg (8.8 lb) in the last few weeks.
8.Patient started drinking a glass of wine every night instead of eating
dinner.
9.Patient has no significant past medical history.
10.Patient is not on any medications.

Question from the doctor:  "What time do you usually go to bed at night?"

Which of the above atomic factual statements answer the question?  Select
at most two statements.  If no statement answers the question, simply say
"I cannot answer this question, please do not ask this question again." Do
not provide any analysis, inference, or implications.  Respond by reciting
the matching statements, then convert the selected statements into first
person perspective as if you are the patient but keep the same information.
Generate your answer in this format:

STATEMENTS:
FIRST PERSON:
```

Example LLM Output

```
STATEMENTS: "Patient goes to bed early at night but is unable to fall
asleep."
FIRST PERSON: "I go to bed early at night, but I can't fall asleep."
```

A.2.5 Additional Patient System Variant: Fact-Classify

Atomic fact decomposition prompt is the same as in Appendix A.2.3.
Patient response generation prompt: for each atomic fact, repeat the following prompting process to decide whether it answers the doctor question, and collect a yes/no answer from the model. The final output is the set of atomic facts with a yes response.

A.2.4 额外患者系统变体: Fact-FP

除了本文主论文中提出的三种患者系统变体，我们还探索了另外两种变体：第一人称（Fact-FP）和二元事实分类（Fact-Classify）。Fact-FP 将患者记录分解为原子事实，选择能够回答后续问题的相关事实，并通过将原子陈述转换为第一人称视角来进行回应。原子事实分解的提示与附录 A.2.3 中相同。

患者反应生成提示 – 事实-FP

```
你是一个有症状清单的病人，你的任务是如实回答医生的问题。

以下是关于你的原子事实列表，仅使用此列表中的信息来回答医生的问题。

1. 患者晚上早早上床，但无法入睡。 2. 患者否认在床上时有焦虑或困扰的想法。 3. 患者
早晨早醒，无法再次入睡。 4. 患者变得越来越易怒，感到越来越绝望。 5. 患者在工作中
的注意力和兴趣减退。 6. 患者否认有自杀或死亡的想法。 7. 患者在过去几周体重减轻了
4公斤（8.8磅）。 8. 患者开始每天晚上喝一杯葡萄酒代替晚餐。 9. 患者无显著既往病史。
10. 患者未服用任何药物。

医生的问题：“你通常晚上几点上床睡觉？”

以上原子事实陈述中，哪些回答了这个问题？最多选择两条陈述。如果没有陈述能回答问
题，只需说：“我无法回答这个问题，请不要再问这个问题。”不要提供任何分析、推论
或含义。通过背诵匹配的陈述来回答，然后将所选陈述转换为第一人称的视角，就像您是
患者一样，但保持相同的信息。请按以下格式生成答案：

陈述：第一人称：
```

示例大型语言模型输出

```
陈述："患者晚上早早上床，但无法入睡。" 第一人称："我晚上早早上床，但我无
法入睡。"
```

A.2.5 额外患者系统变体：事实-分类

原子事实分解提示与附录 A.2.3 中相同。患者反应生成提示：对于每个原子事实，重复以下提示过程，以决定它是否回答医生的问题，并从模型中收集是/否答案。最终输出是带有“是”回答的一组原子事实。

Patient Response Generation Prompt – Fact-Classify

```
You are a truthful assistant that understands the patient's information, and
you are trying to answer questions from a medical doctor about the patient.
Your task is to decide whether the given statement answers the question:

STATEMENT: "Patient goes to bed early at night but is unable to fall asleep."
QUESTION: "What time do you usually go to bed at night?"

Does the statement answer the question? Do not provide any analysis,
inference, or implications. Respond with either YES or NO and NOTHING ELSE.
```

Example LLM Output

```
YES
```

A.3 Qualitative Patient System Failure Analysis

This section provides a detailed analysis of failure cases observed in the MEDIQ Patient system variants Direct and Instruct. These examples highlight limitations and areas for potential improvement in Patient response accuracy and relevance.

Patient System Variant: Direct There are two common failures cases observed by the Direct Patient variant. First, it tends to respond with binary yes/no answers. Although the response is factually correct, it lacks the detail required for a comprehensive medical assessment. First, this behavior does not mimic a typical, realistic patient. Second, detailed information about symptoms, their progression, and related factors is crucial to reaching a diagnosis. Simple affirmative or negative responses do not provide enough context or specifics for an expert to make an informed decision about the patient's condition. The conversation history, which is compiled to assist in final diagnostic decisions, becomes less useful if it is filled with uninformative responses.

Example 1:
Atomic Question: Have you experienced any nausea recently?
Atomic Fact (ground truth): Nausea has subsided.
Patient Response: Yes.
Example 2:
Atomic Question: Have you experienced similar episodes in the past?
Atomic Fact (ground truth): The patient has had 2 similar episodes in the last 5 years.
Patient Response: Yes.

The second common error by the Direct Patient system is hallucinations. In the example below, the Patient system's response includes a hallucinated fact—claiming the initiation of insulin therapy, which is not supported by the patient record. This type of error is particularly problematic because hallucinations introduce false information into the patient's medical history, which can lead to inappropriate medical decisions or interventions based on incorrect data. False information regarding medication can have direct negative effects on patient safety, which calls for the development of factually reliable Patient systems.

Example 1:
Atomic Question: Can you describe any recent changes to your medication? Especially anything related to your diabetes?
Atomic Fact (ground truth): The patient is not on any medications.
Patient Response: I started taking insulin last week for my diabetes.

患者反应生成提示 – 事实分类

```
你是一个诚实的助手，能够理解患者的信息，并且你正在尝试回答医生关于患者的问题。
你的任务是判断给定的陈述是否回答了该问题：

陈述：患者晚上很早就上床睡觉，但无法入睡。 问题：你通常晚上几点上床睡觉？

该陈述是否回答了问题？不要提供任何分析、推断或含义。仅回答“是”或“否”，不要回答
其他内容。
```

示例大型语言模型输出

```
YES
```

A.3 定性患者系统故障分析

本节提供了对MEDIQ患者系统变体Direct和Instruct中观察到的失败案例的详细分析。这些例子突出了患者响应准确性和相关性方面的局限性及潜在改进空间。

患者系统变体：直接型 直接型患者变体观察到两种常见的失败情况。首先，它倾向于以二元的“是/否”回答。虽然这些回答在事实上是正确的，但缺乏进行全面医疗评估所需的细节。第一，这种行为并不模仿典型、真实的患者。第二，关于症状、症状进展及相关因素的详细信息对于做出诊断至关重要。简单的肯定或否定回答无法为专家提供足够的背景或具体信息，以便做出对患者状况的明智判断。为了协助最终诊断而汇总的对话历史，如果充满了无信息量的回答，其作用就会大大降低。

示例 1:
原子问题：您最近有感到恶心吗？原子事实（真实情况）：恶心已经缓解。患者回答：是的。
案例2：原子问题：您过去经历过类似的情况吗？原子事实（真实情况）：患者在过去5年中有过2次类似发作。患者回答：是的。

直接患者系统的第二个常见错误是幻觉。在下面的例子中，患者系统的回应包含了一个虚构的事实——声称启动了胰岛素疗法，而患者记录并未支持这一点。这类错误尤其具有问题，因为幻觉会将虚假信息引入患者的病历，这可能导致基于错误数据的不当医疗决策或干预。关于药物的虚假信息可能直接对患者安全产生负面影响，因此需要开发事实可靠的患者系统。

示例 1:
原子问题：您能描述一下您最近在用药方面的任何变化吗？特别是与糖尿病相关的变化？ 原子事实（事实真相）：患者没有服用任何药物。 患者回答：我上周开始为我的糖尿病使用胰岛素。

Patient System Variant: Instruct The response below by the Instruct Patient system erroneously includes instructive content meant for the healthcare provider rather than addressing the question directly from the patient's perspective. This indicates a potential confusion in role-play within the interactive MEDIQ setup when the conversation log is compiled.

Atomic Question: Have you noticed any unusual bruising on your body?
Atomic Fact (ground truth): Patient experiences easy bruising.
Patient Response: Due to the easy bruising experienced by the patient and her family history of similar issues, it is important to ask the patient if she has noticed any unusual bruising on her body.

A.4 Manual Evaluation of Fact-Select

In order to better understand the quality of the Patient System variant used in subsequent Expert evaluations, we perform a manual annotation of the quality of the responses. A simple preference-based annotation is performed, in which the annotators are provided with a patient record, a Expert follow-up question, and two responses generated by two different Patient systems and are asked to select the one that is more factually correct and does not omit information. The annotators are also instructed to reduce their rating on binary Yes/No answers due to formatting requirements of the framework, but they are free to select any option during annotation. We randomly select one question from each interaction that contains at least 3 follow-up questions, and generate responses with all three Patient systems. For each question, two responses are randomly selected and the annotator is asked to indicate which response is of higher quality. Win rate for each Patient system is calculated as the number of times responses from a system is selected over the number of appearances of responses from this system. There is a tie option for if neither or both responses are good, which we exclude when calculating the win rate. 500 questions are annotated and results are shown in Table 3. We can see that there is an obvious advantage of Fact-Select Patient system over the other systems.

	Direct	Instruct	Fact-Select
Win Rate	36.11	37.36	63.77

Table 3: Win rates from manual evaluation of Patient variants.

B Expert Model Prompts

B.1 Initial Assessment

MEDIQ Expert – Initial Assessment Module Prompt

You are a medical doctor answering real-world medical entrance exam questions. Based on your understanding of basic and clinical science, medical knowledge, and mechanisms underlying health, disease, patient care, and modes of therapy, answer the following multiple choice question. Select one correct answer from A to D. Base your answer on the current and standard practices referenced in medical guidelines.

Task: You will be asked to reason through the current patient's information.

A patient comes into the clinic presenting with some basic information:
"A 40-year-old woman presents with difficulty falling asleep, diminished appetite, and tiredness for the past 6 weeks."

患者系统变体：在下面的回答中，患者系统错误地包含了面向医疗提供者的指导内容，而不是直接从患者的角度回答问题。这表明在生成对话记录的互动MEDIQ设置中，角色扮演可能存在混淆。

原子问题：你有没有注意到身体上有任何不寻常的瘀伤？原子事实（基本事实）：患者容易出现瘀伤。患者回应：由于患者及其家族史中经历了类似的情况，重要的是询问患者是否注意到身体上有任何不寻常的瘀伤。

A.4 事实选择的人工评估

为了更好地理解在后续专家评估中使用的患者系统变体的质量，我们对响应的质量进行了人工标注。进行了一种简单的基于偏好的标注，标注者会获得一个患者记录、一条专家跟进问题以及由两个不同患者系统生成的两个响应，并被要求选择其中更符合事实且未遗漏信息的响应。标注者还被指示在二元“是/否”答案上降低评分，因为框架的格式要求，但他们在标注过程中可以自由选择任何选项。我们从每个至少包含三个跟进问题的交互中随机选择一个问题，并使用所有三个患者系统生成响应。对于每个问题，随机选择两个响应，并要求标注者指出哪一个响应质量更高。每个患者系统的获胜率计算为

	直接	指示	事实选择
胜率	36.11	37.36	63.77

表 3：通过人工评估患者变体的获胜率。

B 专家模型提示

B.1 初步评估

MEDIQ 专家 – 初步评估模块提示

你是一名医学博士，正在回答真实世界的医学入学考试问题。根据你对基础和临床科学、医学知识以及健康、疾病、患者护理和治疗方式背后机制的理解，回答以下多项选择题。从A到D中选择一个正确答案。你的答案应基于当前医学指南中参考的标准实践。

任务：您将被要求推理当前患者的信息。

一名患者来到诊所，提供了一些基本信息：“一名40岁的女性出现入睡困难、食欲减退和疲倦，持续了过去的6周。”

MEDIQ Expert – Initial Assessment Module Prompt (cont'd)

Given the information from above, your task is to choose one of four options that best answers the inquiry.
INQUIRY: "Which of the following is the best course of treatment in this patient?"
OPTIONS: "A": "Diazepam", "B": "Paroxetine", "C": "Zolpidem", "D": "Trazodone"

Medical conditions are complex, so you should seek to understand their situations across many features. First, consider which medical specialty is this patient's case; then, consider a list of necessary features a doctor would need to make the right medical judgment; finally, consider whether all necessary information is given in the conversation above. Think step by step, reason about the patient information, the inquiry, and the options. DO NOT provide the answer choice, keep your response to one paragraph.

Example LLM Output

Based on the patient's presentation of difficulty falling asleep, diminished appetite, and tiredness, the most appropriate course of treatment would be to consider prescribing a medication typically used for depression, like paroxetine or trazodone. These medications can help address symptoms of depression, which can manifest as changes in sleep patterns, appetite, and energy levels. Diazepam and zolpidem are not typically indicated for primary treatment of underlying depressive symptoms. Further evaluation, including a proper assessment for depression and consideration of counseling or therapy, may also be beneficial in the overall management of this patient.

B.2 Abstention Module

For this section, all prompts are appended after the prompt and response from Appendix B.1 in a multi-turn conversation manner.

B.2.1 BASIC Abstain

MEDIQ Expert – BASIC Abstention Module Prompt (appended to previous conversation)

Considering factors above, if you are confident to pick an option correctly and factually, respond with the letter choice and NOTHING ELSE. Otherwise, if you are not confident to pick an option and need more information, ask ONE SPECIFIC ATOMIC QUESTION to the patient. The question should be bite-sized, and NOT ask for too much at once. In this case, respond with the atomic question and NOTHING ELSE.

Example LLM Output

"What time do you usually go to bed at night?"

B.2.2 Binary Abstain

MEDIQ Expert – Binary Abstention Module Prompt (appended to previous conversation)

Considering factors above, are you confident to pick the correct option to the inquiry factually using the conversation log? Answer with YES or NO and NOTHING ELSE.

MEDIQ 专家 – 初步评估模块提示（续）

根据上述信息，您的任务是从四个选项中选择最能回答问题的一个。

询问："在以下情况下，哪种治疗方法是最佳的"
耐心吗?
选项: "A": "地西泮", "B": "帕罗西汀", "C": "唑吡坦", "D": "曲唑酮"

医疗状况是复杂的，因此你应该尝试从多方面了解患者的情况。首先，考虑该患者的病例属于哪个医学专科；接着，考虑医生做出正确医疗判断所需的必要信息清单；最后，考虑上述对话中是否提供了所有必要的信息。逐步思考，推理患者信息、询问内容以及各种选项，不要给出答案选项，将你的回答控制在一段文字内。

示例大型语言模型输出

根据患者表现出的入睡困难、食欲下降和疲劳，最适当的治疗方法是考虑开具通常用于治疗抑郁症的药物，如帕罗西汀或曲唑酮。这些药物可以帮助缓解抑郁症状，而抑郁症状可能表现为睡眠模式、食欲和精力水平的变化。地西泮和唑吡坦通常不用于抑郁症状的主要治疗。进一步的评估，包括对抑郁症的适当评估以及考虑咨询或治疗，也可能有助于该患者的整体管理。

B.2 回避模块

在本节中，所有提示都在附录 B.1 的提示和回答之后以多轮对话的方式附加。

B.2.1 基本 弃权

MEDIQ 专家 – 基础弃权模块提示（附加至之前的对话）

考虑到以上因素，如果你有信心正确且事实性地选择一个选项，请只回复该字母选项，别无其他。否则，如果你没有信心选择一个选项且需要更多信息，请向患者提出一个具体的原子问题。问题应简短，不要一次问太多。在这种情况下，只回复该原子问题，别无其他。

示例大型语言模型输出

你通常晚上几点睡觉?

B.2.2 二进制弃权

MEDIQ 专家 – 二进制回避模块提示（附加到之前的对话）

考虑到以上因素，你有信心根据对话记录事实性地选择正确的选项吗？请仅回答是或否，不要说其他。

Example LLM Output

YES

B.2.3 Numerical Abstain

MEDIQ Expert – Numerical Abstention Module Prompt (appended to previous conversation)

```
Considering factors above, what is your confidence score to pick the correct
option to the inquiry factually using the conversation log?  Answer with the
probability as a float from 0.0 to 1.0 and NOTHING ELSE.
```

Example LLM Output

0.7

B.2.4 Scale Abstain

MEDIQ Expert – Scale Abstention Module Prompt (appended to previous conversation)

```
Considering factors above, how confident are you to pick the correct option
to the problem factually using the conversation log?  Choose between the
following ratings:

"Very Confident" - The correct option is supported by all evidence, and there
is enough evidence to eliminate the rest of the answers, so the option can be
confirmed conclusively.
"Somewhat Confident" - I have reasonably enough information to tell that the
correct option is more likely than other options, more information is helpful
to make a conclusive decision.
"Neither Confident or Unconfident" - There are evident supporting the correct
option, but further evidence is needed to be sure which one is the correct
option.
"Somewhat Unconfident" - There are evidence supporting more than one options,
therefore more questions are needed to further distinguish the options.
"Very Unconfident" - There are not enough evidence supporting any of the
options, the likelihood of picking the correct option at this point is near
random guessing.

Answer in the following format:

DECISION: chosen rating from the above list.
```

Example LLM Output

Somewhat Confident

示例大型语言模型输出

YES

B.2.3 数值弃权

MEDIQ 专家 – 数值回避模块提示（附加至之前的对话）

考虑上述因素，你对使用对话记录事实性地选择正确选项的信心评分是多少？请以0.0到1.0之间的浮点数回答，仅此而已。

示例大型语言模型输出

0.7

B.2.4 规模弃权

MEDIQ 专家 – 规模弃权模块提示（附加到之前的对话）

考虑到上述因素，您对根据对话记录事实性地选择正确选项有多大信心？请在以下评分中选择：

“非常自信” - 正确的选项得到了所有证据的支持，而且
有足够的证据排除其余的答案，所以这个选项可以是
已被最终确认。
“有些自信” - 我有足够的信息可以判断
正确的选项比其他选项更可能，更有信息是有帮助的
做出决定性的决定。
“既不自信也不缺乏自信”——有明显的证据支持这是正确的
这是一个选项，但需要更多证据才能确定哪一个是正确的
选项。
“有些不自信”——有证据支持多种选项，
因此需要更多的问题来进一步区分这些选项。
“非常不自信”——没有足够的证据支持任何一方
选项，此时选择正确选项的可能性接近
随机猜测。

按以下格式回答：

决定：从上述列表中选择的评级。

示例大型语言模型输出

有点自信

B.2.5 Rationale Generation

MEDIQ Expert – Rationale Generation Prompt (appended to previous conversation)

```
Considering factors above, are you confident to pick the correct option to
the inquiry factually using the conversation log?  Answer in the following
format:

REASON: a one-sentence explanation of why you are or are not confident.
DECISION: YES or NO.
```

Example LLM Output

REASON: "I am confident in selecting the correct option based on the patient's presentation of symptoms suggestive of depression, which would warrant treatment with medications like paroxetine or trazodone according to standard medical guidelines for managing depressive symptoms."
DECISION: "YES"

B.3 Follow-up Question Generation

MEDIQ Expert – Question Generation Module Prompt (appended to previous conversation)

```
If there are missing features that prevent you from picking a confident
and factual answer to the inquiry, consider which features are not yet
asked about in the conversation log; then, consider which missing feature
is the most important to ask the patient in order to provide the most
helpful information toward a correct medical decision.  You can ask about
any relevant information about the patient's case, such as family history,
tests and exams results, treatments already done, etc.  Consider what are
the common questions asked in the specific subject relating to the patient's
known symptoms, and what the best and most intuitive doctor would ask.  Ask
ONE SPECIFIC ATOMIC QUESTION to address this feature.  The question should
be bite-sized, and NOT ask for too much at once.  Make sure to NOT repeat any
questions from the above conversation log.  Answer in the following format:

ATOMIC QUESTION: the atomic question and NOTHING ELSE.
```

Example LLM Output

"What time do you usually go to bed at night?"

B.4 Information Integration

None-LLM step. Rearrange the atomic question and patient response into the conversation log. Known patient information $\mathcal{K}$:

```
A patient comes into the clinic presenting with some basic information:
"A 40-year-old woman presents with difficulty falling asleep, diminished
appetite, and tiredness for the past 6 weeks."

Conversation log:
Doctor Question:  "What time do you usually go to bed at night?"
Patient Response:  "Patient goes to bed early at night but is unable to fall
asleep."
```


B.2.5 理由生成

MEDIQ 专家 – 理由生成提示（附加到先前的对话）

考虑到上述因素，您是否有信心根据对话记录准确地选择正确的选项来回答查询？请按以下格式回答：

原因：一句话解释你为什么有信心或没有信心。决定：是或否。

示例大型语言模型输出

原因：“根据患者表现出的抑郁症状，我有信心选择正确的选项，这将根据管理抑郁症状的标准医疗指南，需要使用帕罗西汀或曲唑酮等药物进行治疗。” 决定：“是”

B.3 后续问题生成

MEDIQ 专家 – 问题生成模块提示（附加到之前的对话中）

如果缺少某些特征使你无法自信且依据事实地回答问题，请考虑对话记录中哪些特征尚未被提问；然后，考虑哪个缺失的特征是向患者询问以提供最有帮助信息以作出正确医疗决策的最重要问题。你可以询问有关患者病例的任何相关信息，例如家族病史、检查和化验结果、已进行的治疗等。考虑在与患者已知症状相关的特定主题中常见的提问，以及最优秀、最直觉的医生会问什么。提出一个具体的、原子性的问题来涉及此特征。问题应简明，不要一次性问太多内容。确保不要重复上述对话记录中的任何问题。按以下格式回答：

原子问题：原子问题，仅此而已。

示例大型语言模型输出

你通常晚上几点睡觉?

B.4 信息整合

非大型语言模型步骤。将原子问题和患者回答重新排列到对话记录中。已知患者信息 $\mathcal{K}$：

一名患者来到诊所，提供了一些基本信息：“一名40岁的女性，在过去6周里出现入睡困难、食欲减退和疲劳的症状。”

对话记录：医生提问：“你通常晚上几点睡觉？”患者回答：“患者晚上很早就上床，但无法入睡。”

B.5 Decision Making

MEDIQ Expert – Decision Making Module Prompt (appended to previous conversation)

```
Assume that you already have enough information from the above
question-answer pairs to answer the patient inquiry, use the above
information to produce a factual conclusion.  Answer in the following format:

FINAL CHOICE: correct letter choice and NOTHING ELSE.
```

Example LLM Output

D

C Model Version and Compute

Model Specs and Datasets We use 4bit quantization for Llama-2-Chat-70B and Llama-3-Instruct-70B, and 8bit quantization for Llama-2-Chat-7B, Llama-2-Chat-13B, and Llama-3-Instruct-8B to reduce GPU usage and preserve model utility (Huang et al., 2024). For the OpenAI models, we use the `gpt-3.5-turbo-0125` version for GPT-3.5 and the `gpt-4-turbo-2024-04-09` version for GPT-4. Since no training is needed, we use the development set of iMEDQA with GPT-3.5 to improve the Expert system for faster iteration of the abstention methods. Then, we generalize our findings to other LLM and the entire set of iCRAFT-MD to test our best system.

Hyperparameters For both the Patient and Expert systems, we use a temperature of 0.5 and $top_p = 1$ for top p sampling. Another important hyperparameter is the confidence threshold in the Abstention Module that determines when the Expert should stop asking more questions, which we explore in the Analysis section (§ 5.2) using a grid search instead of arbitrarily setting a value.

C.1 Computing Hardware

We use CPU only for the GPT-based experiments, one A40 GPU for the smaller Llama models (7B, 8B, & 13B), and two A40 GPUs for the 70B models. Time duration of the experiments for each model is roughly proportional the average conversation length (number of follow-up questions) of the interaction, whether rationale is generated, and the self-consistency factor. Experiment time ranges from 30 minutes for GPT-3.5-based Expert systems with no rationale generation and no self-consistency, to 7 days for Llama-2-Chat-70B-based Expert systems with rationale generation, self-consistency, and a high confidence threshold (i.e., more questions are asked).

C.2 Statistical Significance Testing

Since it is expensive to perform repeated runs for all experiments, we approximate the confidence interval using a binomial distribution, where p is the accuracy of the model and n is the dataset size. Therefore, the standard deviation can be calculated as:

$$SD_B = \sqrt{\frac{p(1-p)}{n}} \quad (4)$$

As a sanity check, we repeat the set of experiments in Table 2 with GPT-3.5 and iCRAFT-MD to calculate the random seed standard deviation (SD_R) over one trials. We report the Binomial standard deviation for trial #1 (#1–SD_B) and the random seed standard deviation (SD_R) over one

Model	Non-Interactive			Interactive	
	Full	Initial	None	BASIC	BEST
#1	82.1	53.6	29.3	45.0	**59.3**
#2	78.6	55.0	29.3	47.1	**57.1**
#3	78.6	52.9	31.4	43.6	**57.1**
#4	80.0	52.1	27.9	43.6	**60.0**
#5	80.7	52.1	30.0	46.4	**57.1**
#1–SD_B	3.38	4.22	3.86	4.21	4.17
SD_R	1.51	1.19	1.30	1.63	1.39

Table 4: Repeated trials GPT-3.5 on iMEDQA to compare the Binomial SD over one run and random seed SD over five runs, showing that Binomial SD is a reasonable estimation of the confidence interval with limited data.

B.5 决策

MEDIQ 专家 – 决策模块提示（附加到之前的对话）

```
假设你已经从上述问答对中获得了足够的信息来回答患者的咨询，请使用上述信息得出一个
事实性的结论。按以下格式回答：

最终选择：正确的字母选项，仅此而已。
```

示例大型语言模型输出

D

C 模型版本和计算

模型规格和数据集 我们对 Llama-2-Chat-70B 和 Llama-3-Instruct-70B 使用 4 位量化，对 Llama-2-Chat-7B、Llama-2-Chat-13B 和 Llama-3-Instruct-8B 使用 8 位量化，以减少 GPU 使用并保持模型效用（Huang 等，2024）。对于 OpenAI 模型，我们使用 gpt-3.5-turbo-0125 版本的 GPT-3.5，使用 gpt-4-turbo-2024-04-09 版本的 GPT-4。由于无需训练，我们使用 iMEDQA 的开发集与 GPT-3.5 结合，以改进专家系统，从而加快回避方法的迭代。然后，我们将研究结果推广到其他大型语言模型（LLM）及整个 iCRAFT-MD 数据集，以测试我们最优系统。

超参数 对于患者系统和专家系统，我们使用 0.5 的温度和用于 top p 采样的 $top_p = 1$。另一个重要的超参数是放弃模块中的置信度阈值，它决定了专家何时应停止提出更多问题，我们在分析部分（§ 5.2）中使用网格搜索而不是任意设定值来进行探索。

C.1 计算硬件

我们仅在基于 GPT 的实验中使用 CPU，小型 Llama 模型（7B、8B 和 13B）使用一块 A40 GPU，70B 模型使用两块 A40 GPU。每个模型实验的时间大致与交互的平均对话长度（后续问题的数量）、是否生成推理过程以及自洽性因素成正比。实验时间从基于 GPT-3.5 的专家系统（不生成推理过程且无自洽性）的大约 30 分钟，到基于 Llama-2-Chat-70B 的专家系统（生成推理过程、具有自洽性且高置信阈值，即提出更多问题）的大约 7 天不等。

C.2 统计显著性检验

由于对所有实验进行重复运行的成本很高，我们使用二项分布来近似置信区间，其中 p 是模型的准确率，n 是数据集的大小。因此，标准差可以计算为：

$$SD_B = \sqrt{\frac{p(1-p)}{n}} \quad (4)$$

作为理智检查，我们使用 GPT-3.5 和 iCRAFT-MD 重复表 2 中的一系列实验，以计算在一次试验中的随机种子标准差（SD_R）。我们报告二项式stan试验 #1 (#1–SD_B) 的标准偏差以及随机种子标准偏差 (SD_R)

模型	非交互式		互动的		
	Full	初始	None	基础	BEST
#1	82.1	53.6	29.3	45.0	**59.3**
#2	78.6	55.0	29.3	47.1	**57.1**
#3	78.6	52.9	31.4	43.6	**57.1**
#4	80.0	52.1	27.9	43.6	**60.0**
#5	80.7	52.1	30.0	46.4	**57.1**
#1–SD_B	3.38	4.22	3.86	4.21	4.17
SD_R	1.51	1.19	1.30	1.63	1.39

表 4：在 iMEDQA 上对 GPT-3.5 进行重复试验，以比较一次运行的二项式标准差和五次运行的随机种子标准差，显示二项式标准差是有限数据下置信区间的合理估计。

e

trials in Table 4 to show that the Binomial standard deviation is a reasonable estimation of the confidence interval when we only have one run available.

D Detailed Abstention Results

In order to determine the confidence threshold above which the model should proceed to the final answer, we perform a grid search over the iMedQA dataset. The results are shown in Figure 10. Generally, when the threhold becomes higher, the Expert system asks more questions, and the performance also increases. But as the number of questions becomes too high, the performance stagnates, which might be due to the fact that a lot of the questions will become irrelevant and/or repetitive.

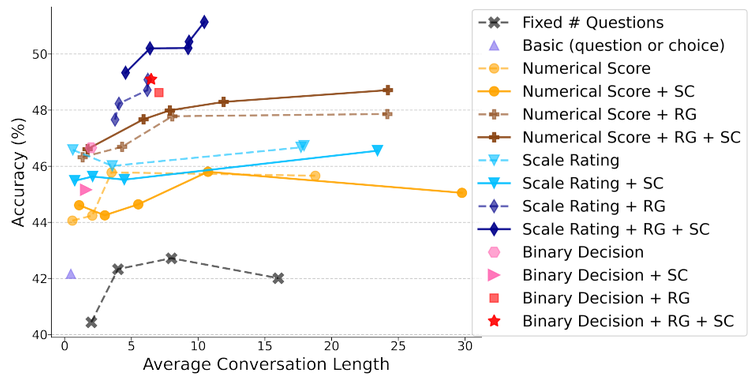


Figure 10: Performance of abstain strategies on iMEDQA. Each line is an abstain strategy with increasing confidence thresholds. Darker colors are results with rationale generation (RG); dashed lines are with self-consistency (SC). The BEST system (Scale+RG+SC,) significantly outperforms the BASIC baseline ( ).

E Results on Additional LLMs

We report the results of Experiment 3.2.1 on the Llama-2-Chat models below. Generally, we observe similar behavior in these models compared to models with higher utility (Llama-3, GPT-3.5, and GPT-4). However, due to the limited model capacity, results for a few settings are less reliable as they are close to at-chance performance (e.g., interactive BASIC for Llama-2-7B).

Task	Model	Non-Interactive			Interactive	
		Full	Initial	None	BASIC	BEST
iMEDQA	**Llama-2-7b**	30.8±1.3	26.3±1.2	26.8±1.2	27.8±1.3	**31.9**±1.3
	Llama-2-13b	37.1±1.4	33.0±1.3	29.9±1.3	31.3±1.3	32.6±1.3
	Llama-2-70b	42.9±1.4	36.7±1.4	31.6±1.3	33.0±1.3	**35.6**±1.3
iCRAFT-MD	**Llama-2-7b**	42.1±4.2	35.0±4.0	30.7±3.9	32.1±3.9	37.1±4.1
	Llama-2-13b	55.0±4.2	42.9±4.2	26.3±3.7	45.7±4.2	38.6±4.1
	Llama-2-70b	60.7±4.1	44.3±4.2	27.9±3.8	45.7±4.2	42.1±4.2

Table 5: Accuracy of Llama-2-Chat models with varying information availability. BASIC is LLM with basic prompting to ask additional questions. Bold results surpass the non-interactive Initial setup.

E.1 Sensitivity to Prompt Variations

LLMs are sensitive to spurious features in the prompt Sclar et al. (2023). We first experiment on various single-turn baselines with prompt variations to evaluate the robustness of the LLaMa-2-chat models Touvron et al. (2023) on the prompt in order to finalize a prompt to standardize future runs. We explore three **system prompts**: Empty (default system prompt), Basic (You

表4中的试验表明，当我们只有一次运行可用时，二项标准差是置信区间的合理估计。

D 详细弃权结果

为了确定模型应在何种置信度阈值以上才能得出最终答案，我们在 iMedQA 数据集上进行了网格搜索。结果如图 10 所示。总体而言，当阈值变高时，专家系统会提出更多问题，同时性能也会提高。但当问题数量过多时，性能会停滞不前，这可能是因为大量问题变得无关或重复。

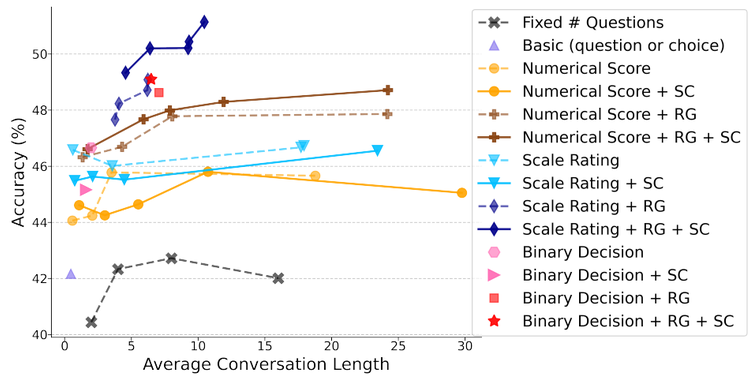


图10：在 iMEDQA 上弃权策略的表现。每条线代表一种具有递增置信度阈值的弃权策略。较深的颜色表示带有推理生成（RG）的结果；虚线表示带有自洽性（SC）的结果。最佳系统（Scale+RG+SC，）明显优于基本基线（）。

E 在其他大型语言模型上的结果

我们在下文报告了对 Llama-2-Chat 模型进行的实验 3.2.1 的结果。总体而言，我们观察到这些模型的表现与更高效用的模型（Llama-3、GPT-3.5 和 GPT-4）相似。然而，由于模型容量有限，少数设置的结果不太可靠，因为它们接近随机表现（例如，Llama-2-7B 的交互式 BASIC）。

Task	模型	非交互式		互动的		
		Full	初始	None	基础	BEST
iMEDQA	**羊驼-2-7b**	30.8±1.3	26.3±1.2	26.8±1.2	27.8±1.3	**31.9**±1.3
	羊驼-2-13b	37.1±1.4	33.0±1.3	29.9±1.3	31.3±1.3	32.6±1.3
	羊驼-2-70b	42.9±1.4	36.7±1.4	31.6±1.3	33.0±1.3	**35.6**±1.3
iCRAFT-MD	**羊驼-2-7b**	42.1±4.2	35.0±4.0	30.7±3.9	32.1±3.9	37.1±4.1
	羊驼-2-13b	55.0±4.2	42.9±4.2	26.3±3.7	45.7±4.2	38.6±4.1
	羊驼-2-70b	60.7±4.1	44.3±4.2	27.9±3.8	45.7±4.2	42.1±4.2

表 5：Llama-2-Chat 模型在不同信息可用性下的准确性。BASIC 是使用基本提示的 LLM，用于提出额外问题。加粗结果超过非互动的初始设置。

E.1 对提示变体的敏感性

LLMs 对提示中的虚假特征很敏感（Sclar 等，2023）。我们首先对各种单轮基线进行实验，并采用提示变体来评估 LLaMa-2-chat 模型（Touvron 等，2023）在提示上的稳健性，以最终确定一个提示来标准化未来的运行。我们探索了三种系统提示：空（默认系统提示）、基本（你

`are a helpful medical assistant`), and the Meditron Prompt adopted from Chen et al. (2023). We also explore three **response prompts** (in the user message) for each combination of model size and system prompt: Answer Only (`Respond with the correct option and nothing else`), Rationale (`Explain the rationale, then select the correct option`), and Permutate (shuffling the option–answer pairs to remove potential dataset bias).

Results on the development set, depicted in Figure 6, show language models, particularly the Llama-2-Chat series, display consistency despite prompt variations. Performance modestly increases with rationale generation, and is unaffected by answer pair shuffling.

Model Size	System Prompt	Response Prompt	Full Context	Basic Info	Question Only
LLaMa-2 7b	Empty	Answer Only	33.04	32.10	31.94
		Rationale	35.09	33.83	30.84
		Shuffle	33.04	29.90	30.84
	Basic	Answer Only	32.49	33.12	28.48
	Meditron	Answer Only	30.81	26.83	26.28
		Rationale	36.11	31.86	29.66
		Shuffle	30.02	26.72	28.64
LLaMa-2 13b	Empty	Answer Only	38.00	36.03	31.63
		Shuffle	37.14	34.23	30.53
	Basic	Answer Only	37.77	35.25	31.00
	Meditron	Answer Only	37.07	33.01	29.93
		Shuffle	37.84	33.67	31.86
LLaMa-2 70b	Empty	Answer Only	43.87	36.66	33.88
	Basic	Answer Only	41.92	35.96	33.10
	Meditron	Answer Only	42.88	36.69	31.58
		Shuffle	42.20	35.92	30.95
GPT-3.5	Meditron	Answer Only	53.42	43.12	35.72

Table 6: Prompt Variations & Single-Turn Baselines.

F How does prior knowledge influence model choice?

General-purposed LLMs tend to provide the most general answers. We hypothesize that this is because the model does not have enough information provided in the context, so it needs to resort to its parametric knowledge and chooses the most common option to maximize the likelihood of providing the correct answer. Therefore, when given more information in the context, the model should rely less on its parametric knowledge and customize to the patient. We quantitatively show this trend by obtaining the most common option according the Expert and calculate the *generality agreement*—percent agreement between the model choice and its belief of the most common option. Selecting the most common option regardless of context yields the correct response only 33.7% of the time, emphasizing the need for the model to focus on provided context rather than prior assumptions from its parametric knowledge. Results show that in setups with increasing information availability (**None**, **Initial**, **Full**), generality agreement shows an decreasing pattern: 50.0%, 40.2%, and 43.4%, respectively.

G How does the nature of the question influence model behavior?

In this section, we show that the nature of the question, such as medical specialty and difficulty, impacts the Expert's interactive behavior. Overall, interaction benefits more difficult questions and certain specialties such as ophthalmology. Figure 11 shows the impact of interactions on diagnostic accuracy across various medical specialties and demographics. Notably, Ophthalmology showed a significant improvement in accuracy from 18.2% to 45.5% after interaction (Figure 12), highlighting the model's potential in specialties with initially low accuracy. Similar trends are evident

是一个有帮助的医疗助手），以及采用自 Chen 等人（2023 年）的 Meditron 提示。我们还为每种模型大小和系统提示的组合探索了三种响应提示（在用户消息中）：仅回答（只回答正确选项，不包含其他内容）、推理（解释推理，然后选择正确选项）、以及排列（打乱选项–答案对以消除潜在的数据集偏差）。

开发集上的结果，如图6所示，显示语言模型，尤其是Llama-2-Chat系列，尽管提示有所变化，仍表现出一致性。生成推理略微提高了性能，而答案对顺序的打乱则没有影响。

模型 Size	系统 提示	回应 提示	Full 上下文	基础 Info	问题 Only
LLaMa-2 7b	空	仅回答	33.04	32.10	31.94
		理由	35.09	33.83	30.84
		洗牌	33.04	29.90	30.84
	基础	仅回答	32.49	33.12	28.48
	美地通	仅回答	30.81	26.83	26.28
		理由	36.11	31.86	29.66
		洗牌	30.02	26.72	28.64
LLaMa-2 13b	空	仅回答	38.00	36.03	31.63
		洗牌	37.14	34.23	30.53
	基础	仅回答	37.77	35.25	31.00
	冥想器	仅回答	37.07	33.01	29.93
		洗牌	37.84	33.67	31.86
LLaMa-2 70b	空	仅回答	43.87	36.66	33.88
	基础	仅回答	41.92	35.96	33.10
	冥想器	仅回答	42.88	36.69	31.58
		洗牌	42.20	35.92	30.95
GPT-3.5	美地通	仅回答	53.42	43.12	35.72

表6：提示变体与单轮基线。

F 先前的知识如何影响模型选择?

通用型大型语言模型（LLM）往往提供最一般性的答案。我们假设这是因为模型在上下文中没有足够的信息，因此它需要依赖其参数化知识，并选择最常见的选项以最大化提供正确答案的可能性。因此，当上下文中提供更多信息时，模型应减少对其参数化知识的依赖，而更加针对个体化情况定制答案。我们通过根据专家获得的最常见选项并计算普遍性一致性——模型选择与其认为最常见选项之间的百分比一致性——来定量地展示这一趋势。无论上下文如何，总是选择最常见的选项，正确响应的概率仅为33.7%，这凸显了模型需要关注提供的上下文，而不是依赖其参数化知识的先验假设。结果显示，在信息可用性逐渐增加的设置下（无，初始，完整），普遍性

G 问题的性质如何影响模型的行为?

在本节中，我们展示了问题的性质，如医学专科和难度，会影响专家的互动行为。总体而言，互动对更难的问题和某些专科（如眼科学）更有益。图11显示了互动对各医学专科和人口统计的诊断准确性的影响。值得注意的是，眼科学在互动后准确率显著提高，从18.2%提升至45.5%（图12），凸显了模型在初始准确率较低的专科中的潜力。类似的趋势也是显而易见的。

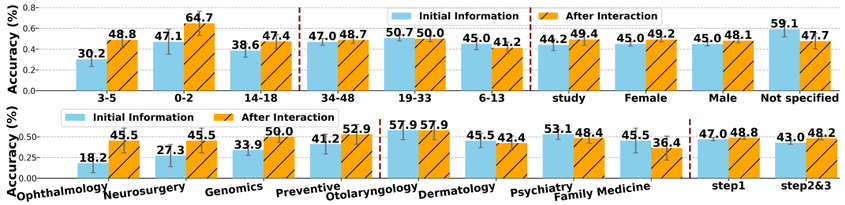


Figure 11: Impact of interactive system on diagnostic accuracy across medical specialties and demographics. Specialties benefiting the most (left) and least (middle) from interaction; improvement by difficulty (right).

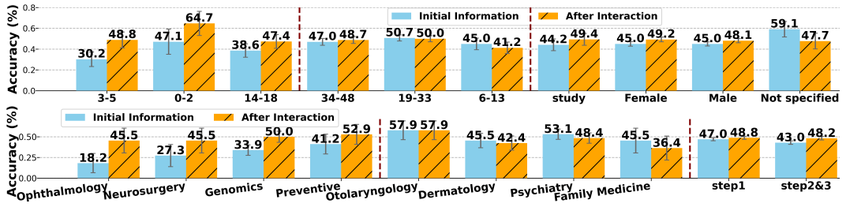
in neurosurgery and genomics, suggesting that interactive information-seeking can enhance decision-making. However, interactions in specialties such as family medicine and psychiatry sometimes result in confusion. The effect of interaction on accuracy also extends across different difficulty levels of medical inquiries. For questions that are more challenging and clinically focused (Step 2 & 3), accuracy improved from 43.4% to 48.3%. Overall, this analysis highlights the importance of tailoring interactive diagnostic systems to specific specialties and types of questions for the maximal benefit.

Patient Information: A 47-year-old man presents to his ophthalmologist for abnormal vision. relevant irrelevant correct option wrong option
Question: Which of the following is the most likely diagnosis?
Options: (A) Anterior uveitis, (B) Normal post-operative visual adaptations, (C) Seasonal allergies, (D) Sympathetic ophthalmia
Initial diagnosis: (A) Anterior uveitis

Doctor Question: Have you experienced any eye pain or redness along with your abnormal vision?
Patient Response: Patient underwent enucleation of right globe and exploratory abdominal laparotomy. Patient's left eye appears red and irritated.
Intermediate diagnosis: (D) Sympathetic ophthalmia

Doctor Question: Have you noticed any changes in your vision, such as blurriness or sensitivity to light, in the affected eye?
Patient Response: Patient has noticed decreased vision in left eye. Patient's left eye appears red and irritated.
Final diagnosis: (D) Sympathetic ophthalmia

Figure 12: Example successful interaction in Ophthalmology.



Fig图 11：交互式系统对各医学专科和人口统计诊断准确性的影响 s.
Spe最受互动益处的专科（左）和最少受益的专科（中）；按难度的改进（右）。

在神经外科和基因组学中，研究表明互动式信息寻求可以增强决策能力。然而，在家庭医学和精神病学等专科中的互动有时会导致混淆。互动对准确性的影响也延伸到不同难度级别的医学问题。对于更具挑战性且临床导向的问题（Step 2和Step 3），准确率从43.4%提高到48.3%。总体而言，这一分析强调了为了最大化效益，应将互动诊断系统针对特定专科和问题类型进行定制的重要性。

Patient Information: A 47-year-old man presents to his ophthalmologist for abnormal vision. relevant irrelevant correct option wrong option
Question: Which of the following is the most likely diagnosis?
Options: (A) Anterior uveitis, (B) Normal post-operative visual adaptations, (C) Seasonal allergies, (D) Sympathetic ophthalmia
Initial diagnosis: (A) Anterior uveitis

Doctor Question: Have you experienced any eye pain or redness along with your abnormal vision?
Patient Response: Patient underwent enucleation of right globe and exploratory abdominal laparotomy. Patient's left eye appears red and irritated.
Intermediate diagnosis: (D) Sympathetic ophthalmia

Doctor Question: Have you noticed any changes in your vision, such as blurriness or sensitivity to light, in the affected eye?
Patient Response: Patient has noticed decreased vision in left eye. Patient's left eye appears red and irritated.
Final diagnosis: (D) Sympathetic ophthalmia

图12：眼科中成功互动的示例。